GEMS-Taiwan 2026

BUSINESS PLAN



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1. Overview

1.1 Climate Stresses & Vulnerability of Agriculture

In a world of climate change, where seasons shift dramatically, temperatures fluctuate uncontrollably, soil salinity escalates, and fungal outbreaks intensify, humanity is forced to choose between immediate technological adaptation or multi-system crop failure in modern agriculture. Abiotic stresses, primarily heat and soil salinity, already cause an estimated 50% of worldwide crop yield losses annually, and climate change further exacerbates the severity and frequency of these issues (Bray et al., 2000; Mittler, 2002). For example, a 1 degrees Celsius rise in average growing season temperatures reduces rice yield by 6.2% (Lyman et al., 2013). Additionally, over 20% of the world's irrigated farmland is already significantly affected by salinity, with an additional 3 hectares becoming unproductive every minute (Zhao et al., 2017). Small farmers produce approximately 70-80% of the global food supply, a proportion that rises to over 80% of all consumed food in many developing sub-regions (Ricciardi et al., 2018). The focus on small farmers is critical because they represent the backbone of local food sovereignty. In East Asia, including Taiwan, where over 90% of agricultural holdings are small-scale family operations averaging less than 1 hectare, these small farmers directly supply more than 80% of domestic fresh vegetables and fruits (Ministry of Agriculture, 2024). In Taiwan's Dahu area, which produces about 90% of the country's strawberries, rising temperatures and typhoon rainfall have been linked to lower first-period yields (Chang, 2020), while year-round tomato production in the subtropical lowlands is challenged by summer heat and a growing risk of salinity stress, with yield strongly suppressed even by short-term mild salinity in a highly heat-tolerant cultivar (Liu et al., 2014). For all the smallholder farmers that make up over 500 million smallholder farms, these staggering numbers mean that a single stress event can potentially wipe out an entire season's income with no recovery mechanism or backup plan.

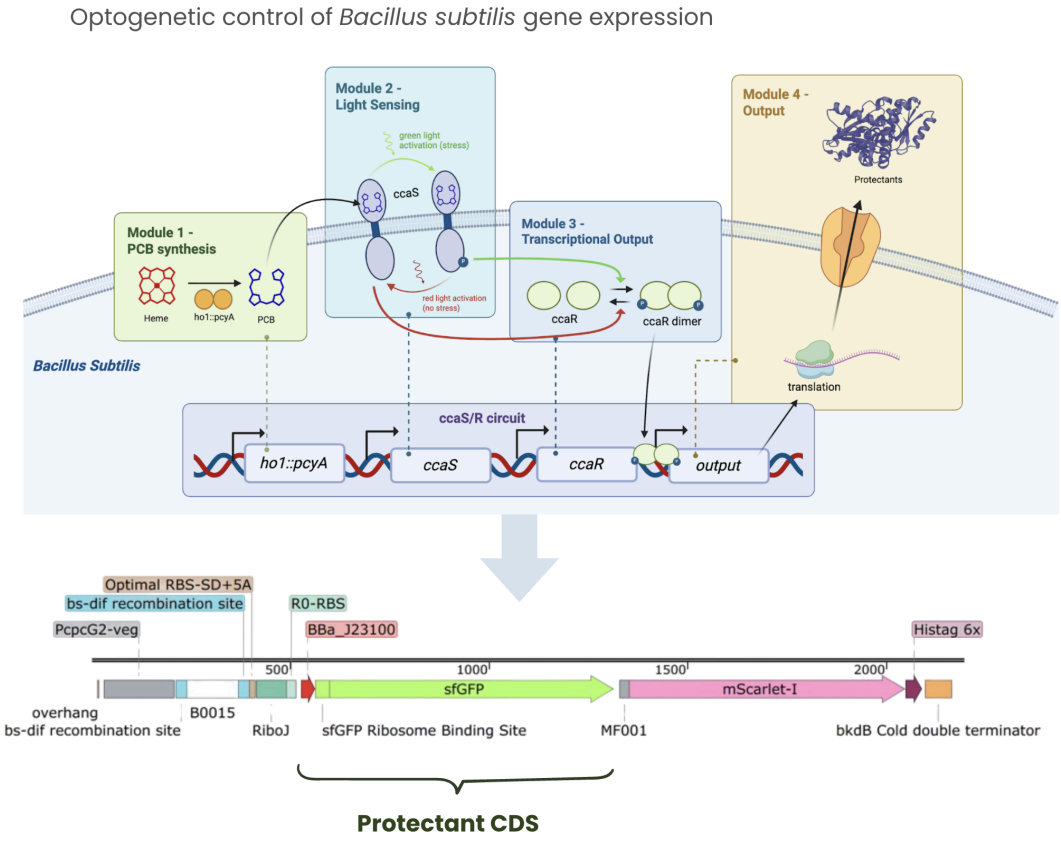
One of the key issues that make this crisis particularly acute is the combination of heat and salinity stress. Under field conditions, these stresses occur repeatedly, compounding cellular damage in ways that can't be triggered by either stress independently. At the same time, plants that are weakened by stress can become vulnerable to pathogens, meaning that even a minuscule instance of salinity or heat can initiate a disease outbreak. Current agricultural systems are not equipped to handle such crises—either current solutions damage soil health and burden farmers with costs they can't afford, or these solutions are reactive and too slow, only able to be applied after visible damage has occurred. Chemical solutions such as fungicides (e.g., mancozeb, azoxystrobin, or copper-based sprays) and synthetic fertilizers remain the most accessible and affordable remedies, but they only further destroy the conditions of the already stressed plants because they are only broadly applied after visible and often irreversible crop damage has already occurred.

1.2 Our Solution

Solutions in the status quo fail to address this devastating reality for farmers. Fertilizers and chemical fungicides remain the most accessible method to address the problems of biotic and abiotic stress, but they are applied broadly—either before the stress occurs, without knowing if or when it will happen. This directly causes overuse,

accelerating soil degradation and increasing input costs that further plunge smallholder farmers into financial crises. Genetic crop modification is another method that is being developed in the status quo, but developing and deploying stress-tolerant varieties takes decades of research and regulatory approval, making it structurally way too slow to combat the predicaments faced in the agricultural sector. Current mechanical as well as physical interventions such as shade structures and irrigation systems may provide broad information about environmental conditions but generally require farmers to manually interpret the data to determine appropriate interventions. By integrating digital twins with AIoT sensors, we can predict when crop stress will occur, allowing farmers to act before problems develop instead of relying on broad treatments that may not necessarily work. Through our genetically modified solution, we are able to double the fresh weight growth of Arabidopsis by applying our ACCD protectant as a treatment in comparison to a control treatment using TB medium. By applying the same treatment process to other plants, we are hopeful about similar results and similarly increasing plant yield.

Compared to current commercial biostimulant manufacturers, such as CH Biotech with its plant growth regulators or peptide/amino acid formulations, our solution occupies a distinct operational niche. While commercial formulations cannot dynamically adapt when a sudden heatwave or salinity surge transitions into an outbreak, our solution provides on-site and on-demand biomanufacturing. By placing a localized bioreactor system directly at farms, active bioprotectants can be produced on-site only when environmental sensors detect real-time stress. This on-demand approach eliminates current issues such as shelf-life degradation of protectants, avoids unnecessary chemical application, and develops fresh, highly potent proteins to crops immediately before irreversible cellular damage occurs.



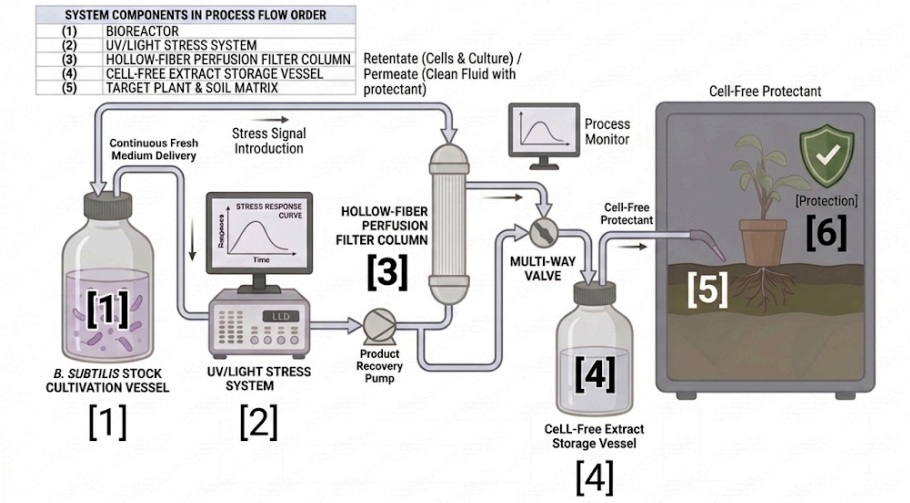
Our solution utilizes an optogenetic control mechanism centered around the two-component CcaS-CcaR system in *Bacillus subtilis*. The CcaS-CcaR system enables precise, non-invasive regulation: when environmental sensors inside the bioreactor detect stress, the system transmits light signals to seamlessly decouple bacterial cell growth from protectant production while ensuring that the bacteria itself doesn't escape the bioreactor (Castillo-Hair et al., 2019). A modular manufacturing design is essential to allow an engineered chassis to easily swap genetic expression cassettes, enabling on-demand production of diverse, customized bioprotectants tailored to specific biotic and abiotic threats.

As climate-related stress events continue to escalate, there is a growing need for agricultural systems that can both detect stress conditions and automatically provide targeted protection before irreversible crop damage occurs.

1.3 Mission

Our mission is to strengthen agricultural resilience by increasing crop yield caused by combined biotic and abiotic stress while decreasing dependence on chemical interventions. Through our experimentation on Arabidopsis, we have found that our genetically modified ACCD is able to double the fresh weight of crops when treated with treatment in comparison to control treatment in TB medium. We strive to be able to apply this to other crops and yield similar results. As climate change continues to intensify environmental stressors such as heat, waves, salinity, drought, and irregular weather patterns, farmers are increasingly challenged by conditions that directly threaten crop productivity and food security. These abiotic stresses often weaken plants, making them more susceptible to

fungal pathogens and other biological threats. Through the development of our bioreactor, which will release a protectant to plants in times of stress, we aim to provide farmers with a more sustainable and proactive alternative. Rather than responding after visible damage has already occurred, our approach focuses on detecting stress conditions early, deploying our protectant only when necessary, and doing so in precise amounts. By reducing unnecessary chemical applications while improving crop resilience, we hope to help farmers maintain stable yields while minimizing environmental impact. To achieve this, our project incorporates three core engineering aspects.



First, we incorporate an optogenetic control mechanism, utilizing the photoreversible CcaS-CcaR system that simultaneously regulates bacterial cell growth and controls protectant synthesis. By utilizing light as a non-invasive switch inside our chassis host, *Bacillus subtilis*, we decouple biomass accumulation from protein expression, allowing the bacteria to reach high cell densities before the green light triggers a rapid, high-yield supply of protectants. Second, our system employs true on-site and on-demand biomanufacturing. Producing protective proteins directly at the farm level precisely when environmental sensors detect real-time stress eliminates the problem of transport delays, rapid shelf-life degradation, and retail markups associated with centralized commercial biostimulants. To ensure complete biosecurity, our sealed bioreactor employs a hollow-fiber membrane that prevents the *Bacillus subtilis* from escaping, physically containing the bacterial cells within our bioreactor while releasing only the protectants onto crops. Third, our modular manufacturing design allows engineered organisms and cassettes to be rapidly swapped, producing customized protectants tailored to the specific stress combinations faced by different plants. Central to our mission is achieving environmental sustainability and long-term economic stability for smallholder farmers. We ensure precise, targeted control of protectant usage, shifting to localized, on-site production to reduce the carbon footprint generated by long-distance transport logistics. Furthermore, by replacing chemical fertilizers with biological protectants, we prevent soil microbiome degradation, protect local water systems, and decrease agricultural chemical pollution.

Ultimately, our mission extends far beyond just improving agricultural productivity. We hope to support the livelihoods of climate-vulnerable farming communities, promote sustainable food production, and address some of the most pressing challenges that global agriculture is currently facing.

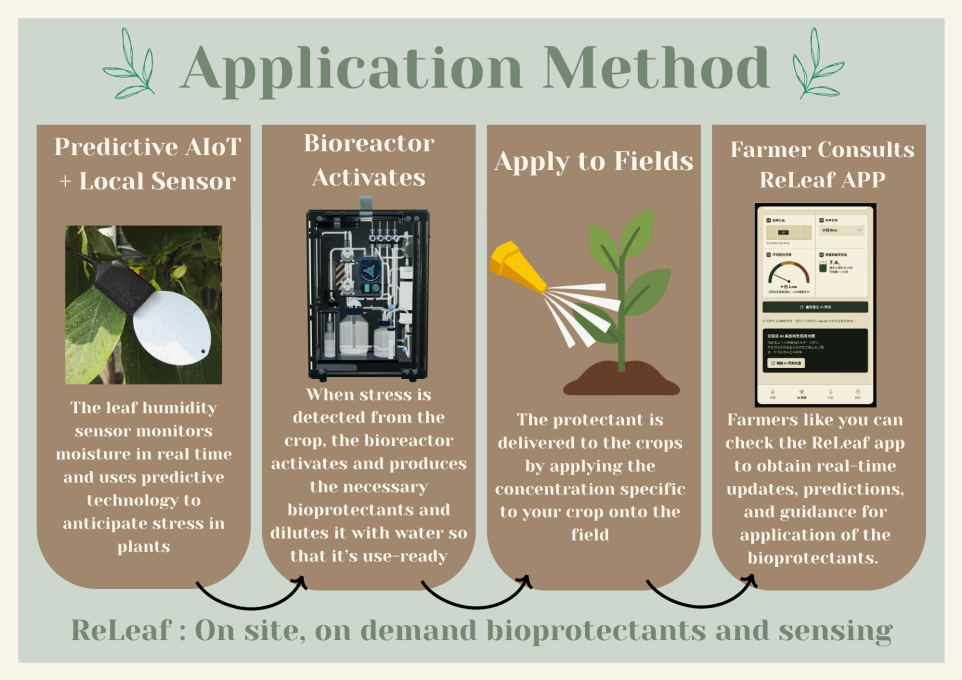
1.4 Vision

Our vision extends far beyond the scope of iGEM, aiming to pioneer a shift toward climate-resilient agriculture. To transition from our proof-of-concept to a more practical, field-ready solution, our future development will focus on scaling and refining our hardware platform. We envision a self-contained, fully automated, low-cost microbioreactor unit designed specifically to be integrated into farms. By combining our light-regulated protectant synthesis in *Bacillus subtilis* with AI stress detection and forecasting, we envision agricultural systems that operate independently of reactive, post-damage management practices. To achieve this transition and realize our long-term commercial goals, our product strategy follows a five-step execution framework:

1. Field Trials for Target Crops: We will begin with controlled field trials on target vegetable crops by applying extracted peptide solutions directly. Once the efficacy of our solution is confirmed, our testing will then transition to deployment of our bioreactor solution in open-soil environments to validate real-world performance under field conditions. Furthermore, we will include a droplet sensor in our bioreactor to minimize leakage of our protectant during production.
2. Certification & Regulatory Clearance: We will navigate the necessary regulatory pathways for product registration and safety certification. By proving absolute containment via our 0.22 um hollow fiber membrane filtration system, our cell-free delivery mechanism circumvents the regulations associated with open GMO releases, qualifying our output as a plant biostimulant
3. Global Partnerships & Market Expansion: Building on initial field success in Taiwan, we will establish strategic partnerships with local agricultural cooperatives, research institutions, and international distributors. These collaborations will serve as a launchpad to expand into high-demand regional markets in Southeast Asia, eventually spilling over into global markets across Europe and the United States.
4. Bridging the Gap Between Young & Old Farmers: Central to our mission is ensuring that every farmer has their own bioreactor system. By pairing an intuitive, zero-configuration physical hardware interface with AI-driven stress detection and predictive forecasting, our system automates the preventive delivery of protectants. This significantly reduces the reliance on traditional, hard-earned farming experience, making crop protection accessible, lowering operational barriers for younger generations entering agriculture, and shielding aging farming communities from catastrophic climate-driven crop losses.
5. Automation to Address Agricultural Labor Shortages: Labor shortages are a structural problem in Taiwan's agriculture. The number of people working in agriculture, forestry, fishery, and animal husbandry fell from about 1.3 million in the 1980s to about 542,000 in 2021 (Yang et al., 2022). Roughly 49% of vegetable farms reported severe labor shortages, and small-scale farming makes full mechanization difficult (Han & Yang, 2017). The Council of Agriculture itself acknowledges that farm labor demand cannot be reduced without automation (Han & Yang, 2017). The Netherlands shows this kind of automation is already practical. Self-driving spraying robots are the most common robot in Dutch greenhouses (Claver, 2019), and Dutch farms run automated on-site microbial fermentors that need only about five minutes of manual work a day (Pig Business, n.d.), as cited in Pig. The remaining barriers are time and cost, not feasibility. Therefore, our bioreactor follows this same logic. Sensing, stress forecasting, protectant production, and dosing run without a farmer monitoring or intervening, so protection no longer competes with planting and harvesting for scarce labor. (Claver, 2019)

2. Product Description

2.1 Our products

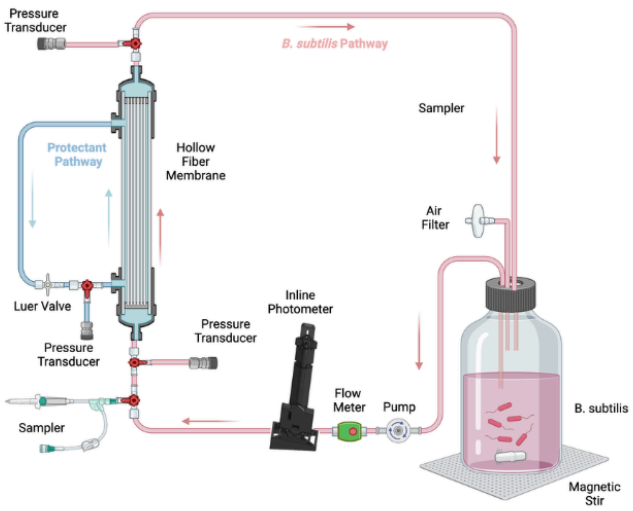


Our product is a field-deployable, three-part system. First, users will be able to purchase a sealed bioreactor that contains our engineered *Bacillus subtilis* that produces and delivers targeted biological protectants the moment environmental sensors detect crop stress. A hollow-fiber membrane keeps all living bacteria contained inside the device, releasing only protectant molecules like LEA14 peptides and the antifungal defensins to the crop. This cell-free delivery approach lets the system avoid the regulatory and trust barriers conventional GMOs face in the field.

An optogenetic CcaS-CcaR system uses green and red light to decouple bacterial growth from protectant production, letting the culture reach high density before rapidly shifting into output once stress is detected. A modular cassette design allows the same hardware to produce different protectants for different stress combinations. This gives farmers one adaptable platform instead of a single-purpose product.

Compared to fixed-schedule sprays or centralized biostimulants, ReLEAF offers on-site, on-demand production that responds to real conditions in real time. This targets smallholder farmers, who grow 70 to 80 percent of the world's food yet are most exposed to a single bad stress event. The result is a proactive, sustainable alternative to reactive chemical interventions.

2.1.1 Bioreactor

Product picture	 The bioreactor is a compact optogenetic bioreactor system using *Bacillus subtilis* to produce protective compounds (LEA proteins, ACCD, and BoPep4) for management of different types of crop stress. Light-controlled gene expression regulates compound production in real-time, delivering targeted protection to smallholder farms. - customizable, adaptable - precise, on-site, on-demand - automated
Price	USD $2,280
Description	Our field-deployable bioreactor is a sealed system containing engineered *Bacillus subtilis* that uses a hollow-fiber membrane filtration system to deliver protectant molecules directly onto the plant leaf zone of vegetable crops. By restricting all living bacteria inside the device while allowing only the secreted protectants to exit, it automates crop defense without the need for manual spraying or application. The device functions with minimal farmer intervention, requiring only occasional maintenance checks or a cartridge replacement when the protectant runs low. It is fully automated by an Arduino-based control unit, a circulation pump, and a green and red LED optogenetic control system. This integrated tech stack continuously monitors environmental conditions and automatically activates specific stress-response modules the moment predefined thresholds are breached.
Storage	The bioreactor should be stored indoors at room temperature in a clean, dry space away from direct sunlight. Before each use, check for visible damage or contamination, and replace tubing if cracked or cloudy. The recommended storage temperature, shelf life, and post-opening conditions will be set

2.1.2 Sensor and AIoT System

Product picture	 - direct field-to-bioreactor linking - early stress detection
Price	USD $125
Description	The leaf humidity sensor sits on the leaves of crops and tells us the humidity of the leaf surface. This is important as leaf humidity tends to be very different from the humidity reported at weather stations. When the leaf is dry and the weather is very warm, the plant would be rapidly losing water, which is usually the first sign of drought or heat stress. The sensor then sends its readings to our AIoT system, which uses the readings to forecast when the crop is going to need protection. By using the sensor data, the AIoT system also predicts upcoming weather conditions, while the sensor tells the user how the crop is doing. If the leaves show signs of stress before the forecast predicted, the AIoT system can bring production forward accordingly. The AIoT translates this multi-source data into guidance for farmers without technical background understanding of the bioreactor. Together, the sensor and AIoT link conditions in the field directly to production in the bioreactor, ensuring protectant is available exactly when the crop needs it.

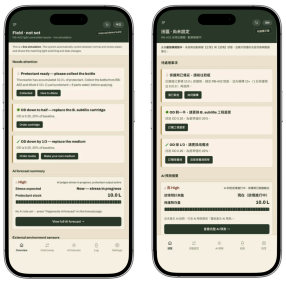
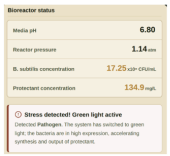
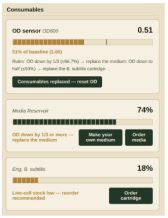
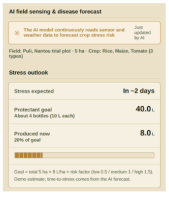
Storage	To keep the sensor clean, store away from direct sunlight. At the end of each growing season, retrieve it from the field. Before redeploying it, verify its occurrence against a reference humidity blank and recalibrate it according to specifications. The recommended storage temperature, shelf life and post-opening conditions will be set through stability testing of the final formulation and packaging before commercial distribution.

2.1.3 Bacteria Culture Cartridge

Product picture	 The ReLEAF bacterial culture cartridge is a sealed supply of Bacillus subtilis for the ReLEAF bioreactor. It serves as the bacterial replenishment system, which helps to maintain the bacterial population and supports continuous protectant production. - Biosafe - Convenient - Durable
Price	USD $90
Description	The bacterial culture replenishment system supplies a sealed *Bacillus subtilis* inoculum in a cartridge that fits the bioreactor's filter and cartridge design. The cartridge drops into the reservoir port from above and opens automatically once it is seated, so the culture goes straight into the reservoir. There is no need to handle an open culture, and the bioreactor stays closed throughout. Each package includes two culture cartridges together with the main unit. Thus, a replacement is on hand from the first season. Since the culture arrives ready to use, specialized workers do not need a lab, special equipment, or any microbiology skills to prepare fresh culture themselves.

Applications	When the in-line photometer measures the culture's optical density (OD600) at 50%, the bioreactor mobile app notifies the farmer and a trained operator that a new cartridge is due. The operator then follows the on-screen instructions on the control screen. Under suitable operating conditions, the bacteria will be able to resume growth and continue producing protectants in the bioreactor.
Storage	To preserve viable bacterial populations, keep the cartridges sealed in a cool, dry environment away from direct sunlight and excessive heat until use. Wear gloves and eye protection when handling cartridges or the cleaning solution. Do not open or force a cartridge outside the machine. The recommended storage temperature, shelf life and post-opening conditions will be set through stability testing of the final formulation and packaging before commercial distribution.

2.1.4 Field Console Mobile App

Product picture	  **AI forecast summary:** Risk level, days until stress, protectant stock and risk per crop.  **Bioreactor status:** Media pH, pressure, B. subtilis and protectant concentration.  **Consumables:** Media and engineered-cartridge levels with ordering buttons.  **Live AI forecast & stress outlook:** Reads sensor and weather data and shows the last update.  **Field Setup (Farmer inputs)** Field location, crops and area, number of sensors and bioreactors installed in field.  **Light-control system diagram:** Red light keeps output off, green light starts protectant production.  **Environment sensors:** Air temperature, humidity, soil moisture, soil EC and light.  **Equipment & Price Consultation:** Calculate the number of equipments needed and added all prices. - Simple - Proactive - Customizable

Price	Free, comes with purchase of our other products
Description	The app pairs up with the bioreactor and consolidates operational information into five modules: Overview, Field Setup, AI Forecast, Log, and Settings. The Overview page shows a prioritized list of required actions, a summary of the current AI forecast, real-time field and reactor measurements, a schematic of the red/green light-control system, and the status of consumables (medium and culture). The Field Setup module records the inputs from farmers regarding the farm location, the crops grown, the area allocated to each, and the numbers of soil sensors and bioreactors already installed. From this data, the system estimates the recommended equipment and its total cost. The AI Forecast module reports a stress risk level for each crop, the estimated time until stress onset based on the farmer's location, the target quantity of protectant, and the quantity produced to date. Orders are collected in a cart accessible from the header. In the present version, all sensor readings are simulated for demonstration purposes.
Applications	The app is intended for farmers who grow one or several crops and need to act before stress damages them, without laboratory skills. A farmer facing seasonal heat, soil salinity, or fungal disease can use the forecast to prepare ahead, then collect the protectant the bioreactor produces, dilute it by the stated factor, and apply it through irrigation or spraying. A farmer who adds crops or land can enter the new areas to see how many sensors and bioreactors are needed, what they cost, and what remains to be bought beyond the installed equipment.

2.2 Technology

Our System:

Our system is based on the CcaSR light-responsive two-component system from a research paper, which has been demonstrated for controlling gene expression in *Bacillus subtilis*. This gives us a solid reference for which our cloning system is based. The first module is PCB synthesis. The CDS of this module is Ho1:pcyA. Ho1 basically converts heme to BV, which is then converted to PCB by pcyA. PCB is essential in this system because it functions as the chromophore required for the light-sensing system. Next is the light-sensing module, where the CcaS photoreceptor detects the light signal. Once activated, CcaS initiates phosphorylation of CcaR, the response regulator. CcaR then acts downstream to regulate transcription of our target gene. Therefore, the overall system converts an external light signal into controlled gene expression. Based on this principle, we designed our downstream construct to express our protectant proteins, which I'll talk about later. This version of the system is used for LPA testing, though, which is when we'll test if our circuit responds to light. In our final product, the output module will be replaced with this construct, which is essentially the same inducible promoter and rbs, just with this sfGFP part that can be replaced by any protectant CDS and secretion peptide the farmer wishes to use. With this, we'll be able to assemble customizable systems. The mScarlet bright fluorescent protein will allow

farmers to visually see that the protectant has been produced, as the plasmid will turn a red-pinkish color when protein is produced. After establishing how we can control gene expression, the next question is, what should we express? Our three main final selections of protectants are ACC deaminase, which targets salinity and drought stress; LEA 14 for osmotic stress; and BoPep4 for salinity stress. We assembled them into constitutive versions of our output module for plant and efficiency testing. The structure is the same, except for the constitutive promoter instead of the inducible, to allow constant gene expression. We made different versions with our three protectants and our secretion peptides. With these protectants, we carried out drylab and wetlab collaborations, consisting of mutations of our protectant CDS. The two main parts were truncation mutations and point mutations. We did truncation mutations because smaller peptides enter plants more efficiently, so if our protectant protein is smaller, then it'll enter plants more easily. As you can see in the three sequences below, it's visibly a lot shorter than the wildtype. On the other hand, point mutations stabilize molecule structure during field application and improve protein efficiency. The third sequence, corresponding to a point mutation sequence of BoPep4, only has one amino acid different from the wild type.

Bioreactor Technological Overview:

The bioreactor produces crop stress protectants at the point of use. Engineered *Bacillus subtilis* grows in a sealed loop and secretes protectants through a hollow fiber membrane, which holds the cells back. The output is a cell-free liquid that feeds straight into an existing hydroponic or irrigation line. Light controls when production happens, so production can be turned on for a stress event and off again after. The living part arrives as a sealed cartridge, so running the machine does not require a lab.

Strengths:

- The organism never leaves the machine, so the product is cell-free, not a GMO release
- Made on-site, so no cold chain, no shelf life, no shipping of a degradable protein
- Light as the input means no inducer cost and no residue in the product
- Platform hardware: a new cartridge means a new protectant and a new crop
- Cartridges give recurring revenue on top of the hardware sale
- Output drops into irrigation lines growers already have
- Off-the-shelf parts keep unit cost and servicing low

Weaknesses:

- Scales by unit count, not unit size, so cost per dose stops dropping
- The grower has to refill cartridges after medium runs out and know how to deal with potential issues
- Small early market, controlled environment, and high-value crops only

The sensors continuously monitor the farmer's field. Leaf humidity sensors track moisture stress on leaves in real time, while our AIoT system (paired with a digital twin, a virtual model of the farm's specific conditions) processes weather data and field conditions to forecast when crops will need protection. The digital twin learns the farmer's

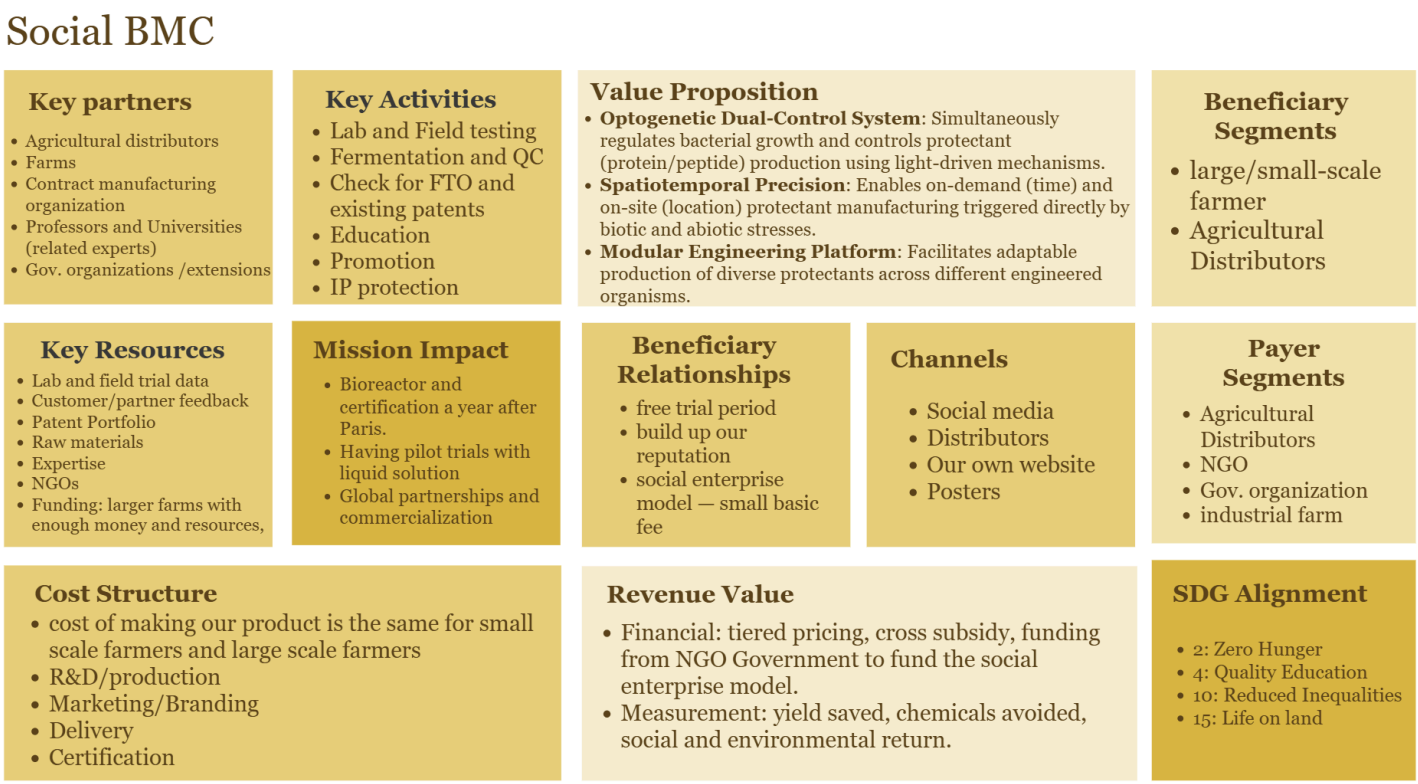
patterns over time and improves its predictions so that the system gets smarter the longer it is used by the farmer. This real-time data feeds directly into the bioreactor, so production can start before stress arrives.

The bioreactor produces crop stress protectants for the farmer. Engineered *Bacillus subtilis* grows in a sealed loop and secretes protectants through a hollow fiber membrane, which holds the cells back. The output is a cell-free liquid that the farmer can then use to spray on their crops. The *Bacillus subtilis* arrives as a sealed cartridge, so running the bioreactor itself does not require the farmers to be in a lab. The app then alerts the farmer of when to refill the growth medium and when to substitute the *B. subtilis.* Maintenance of the bioreactor is considerably minimal since sensors and the bioreactor itself can function indoors when supplied with the prepared medium or *B. Subtillis* cartridges, or farmers can prepare their own, following our protocol.

The app brings everything together on the farmer's phone. It tailors recommendations to each farmer's specific crops and stress conditions. Farmers will be notified when the protectant is being produced, what stress is being forecasted, and how the plants are responding to the protectants. When the protectant is ready, the app will notify farmers with the exact dilution ratio for their crop type and stress level. Real-time updates, historical records, and quick notifications will help farmers make better decisions about their crops based on what's actually happening in the field itself as well as in the bioreactor.

3. Business Model

3.1 Business model canvas



Key Partners

Our operations rely on collaboration with agricultural distributors, farms, and contract manufacturing organizations to bring our solutions to market. We collaborate with professors and university experts for technical guidance, alongside government organizations and extensions to ensure aligned regulatory support and community reach.

Key Activities

Core operations focus on extensive lab and field testing, fermentation, quality control (QC), and verifying Freedom to Operate (FTO) alongside existing patents. Additionally, we actively invest in education, promotional outreach, and intellectual property (IP) protection to safeguard and drive our technology forward.

Value Proposition

Our core value lies in an advanced biological platform featuring an optogenetic dual-control system that simultaneously regulates bacterial growth and drives protective protein/peptide synthesis using light mechanisms. By providing spatiotemporal precision, the platform enables on-demand and on-site manufacturing of protectants directly triggered by surrounding biotic and abiotic environmental stresses. Furthermore, its modular engineering

platform ensures high adaptability, allowing seamless customized protectant production across a variety of engineered organisms.

Beneficiary Segments

The primary beneficiaries targeted by our initiative include both large-scale and small-scale farmers who directly utilize the crop protection solution, as well as agricultural distributors managing local supply networks.

Key Resources

Our critical assets comprise essential lab and field trial data, customer/partner feedback, a growing patent portfolio, quality raw materials, and specialized technical expertise. Funding and support are further grounded through partnerships with NGOs, government organizations, and larger farms equipped with appropriate resources.

Mission Impact

Our roadmap targets establishing a bioreactor setup and securing certification within a year following Paris, alongside running pilot trials using liquid solutions. Long-term impact is driven through expanding global partnerships and commercialization efforts to scale adoption worldwide.

Beneficiary Relationships

We build trust with beneficiaries by offering a free trial period to establish our reputation and demonstrate value. Operating under a social enterprise model, we maintain an accessible, small basic fee structure to keep the solution affordable for underserved growers.

Channels

We reach our audiences through active social media engagement, direct distributor networks, our official website, and targeted print media such as posters in regional agricultural hubs.

Payer Segments

Financial sustainability is maintained by securing revenue and funding from agricultural distributors, NGOs, government organizations, and well-funded industrial farms.

Cost Structure

Our cost drivers consist of core expenses in R&D and production, marketing and branding, delivery logistics, and official certification. Crucially, the cost of manufacturing our product remains identical for both small-scale and large-scale farmers.

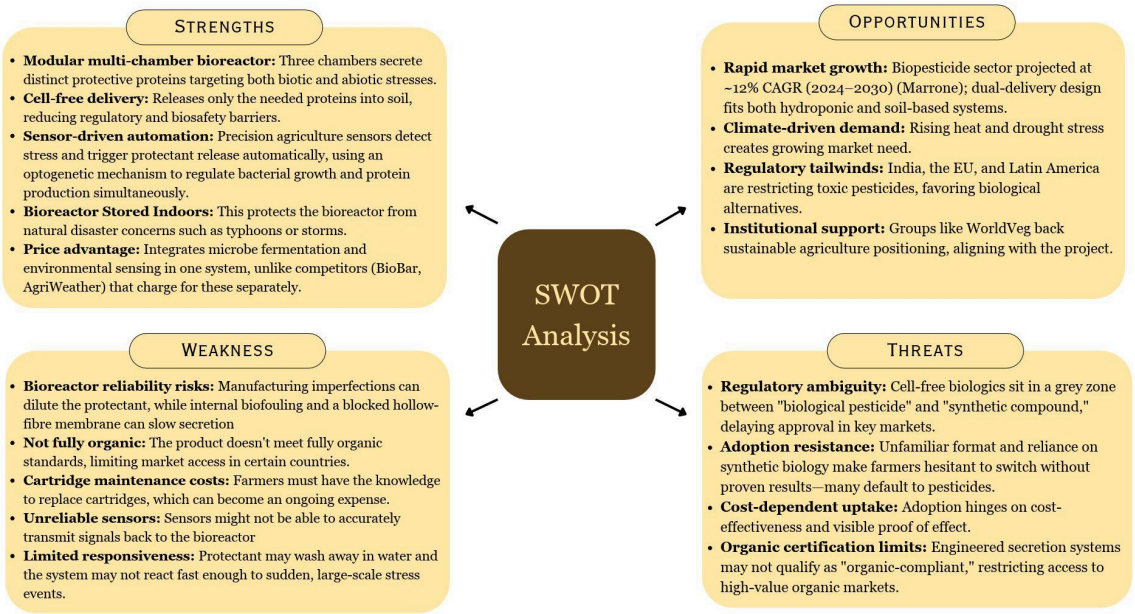
Revenue Value

Financial viability relies on tiered pricing models, cross-subsidies, and direct grant/funding support from NGOs and government bodies to sustain our social enterprise model. Success is measured beyond monetary terms through yield saved, chemical usage avoided, and overall social and environmental returns.

SDG Alignment

Our initiative directly advances key United Nations Sustainable Development Goals, including SDG 2 (Zero Hunger), SDG 8 (Decent Work and Economic Growth), SDG 12 (Responsible Consumption and Production), SDG 13 (Climate Action), and SDG 15 (Life on Land).

3.2 SWOT Analysis



Strengths

Our SWOT analysis highlights the strong potential of our project while weighing key challenges that must be addressed for successful implementation. The system's primary strengths lie in its modular multi-chamber design, which allows for the targeted, light-triggered production of three specific protective proteins (LEA, ACCD, and BoPep4). Additionally, its cell-free delivery approach reduces biosafety and regulatory concerns while its integration of electronic sensors and AI forecasting allows it to deliver precise protectant doses to both soil-based and hydroponic systems. These features provide a sustainable alternative to conventional chemical pesticides or outdated methods while minimizing environmental contamination.

Opportunities

Externally, the project is in a beneficial position due to the rapidly growing biocontrol market, increasing climate-related crop stress, and global support for sustainable agriculture. While climate change continues to intensify the severity of stress on crops, sustainable agriculture is essential. Additionally, many countries strictly

regulate or are in the process of phasing out toxic chemical pesticides, which creates a strong push toward biological alternatives. This widens our market opportunity across both traditional field agriculture and high-tech hydroponic systems, increasing adoption potential amongst growers seeking sustainable solutions. Many companies around the world charge the same amount for microbe fermentation and sensing separately while our product is a two-in-one for the same price. Furthermore, since the Taiwan government is willing to give subsidies to local farmers, it makes our product more accessible to our stakeholders. Also, our product can be tailor-made for different farmers from different backgrounds, which expands our market.

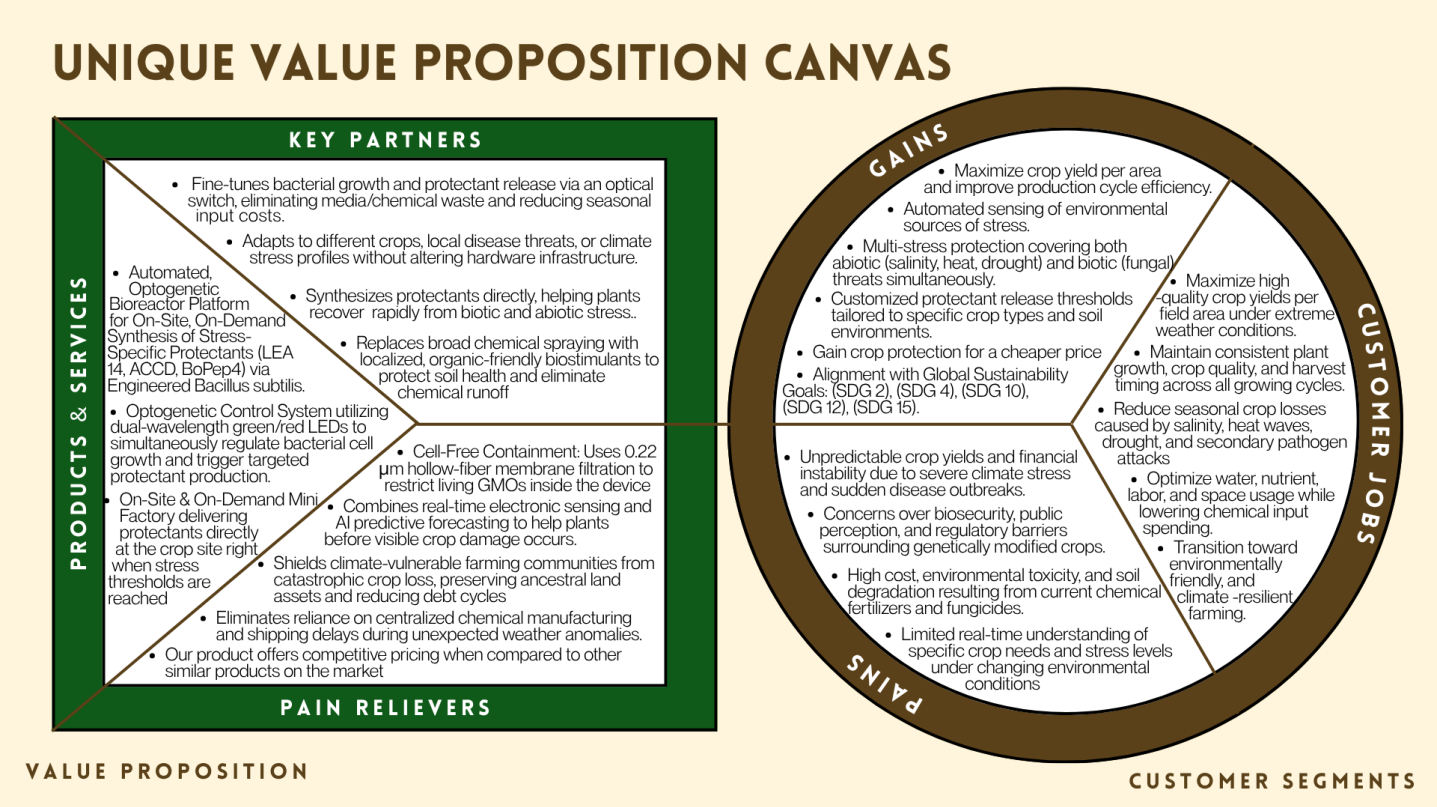
Weaknesses

However, the project also has several weaknesses. There could be potential protectant loss from being washed away by water. Additionally, the bioreactor might not work fast enough when met with a sudden mass onslaught of stress. Imperfections in the bioreactor could lead to dilution of the protectants, while biofouling inside the chambers may reduce protein yield and shorten the system's lifespan. The product may also face limitations in countries with strict organic certification requirements because organic certification standards exclude engineered inputs, so we do not target organic farms. Furthermore, farmers would need to learn how to replace cartridges, which could increase long-term maintenance costs and create an additional barrier to adoption. With the bioreactor, the hollow-fiber membrane might get blocked, which can slow down or potentially prevent excretion.

Threats

Finally, several external threats could affect the success of the product. Unclear regulations surrounding cell-free agricultural biologics may delay approval, especially if the product falls between the classifications of a biological pesticide and a synthetic biology-derived compound. Farmer adoption may also be limited by unfamiliarity with the technology, concerns about cost-effectiveness, and reliance on traditional chemical pesticides. In addition, some farmers may view synthetic biology as too risky, while restrictions on organic certification could limit access to high-value organic farming markets. Lastly, the bioreactor may get destroyed by natural disasters such as the frequent earthquakes or typhoons in Taiwan.

3.3 Value Proposition



Current Challenges and Solutions

Our unique value proposition highlights the advantages of our bioreactor system containing engineered *Bacillus subtilis* that continuously monitors the conditions of plants and releases protectants when predetermined thresholds are reached. Our system offers farmers an automatic, customizable, more effective, and sustainable way to reduce crop losses from both environmental and disease stress while lowering chemical use and improving yield reliability. Current management methods often depend on repeated chemical spraying, manual monitoring, and frequent maintenance, leading to higher labor costs, chemical waste, environmental pollution, and inefficient use of resources. In addition, these traditional solutions are rarely tailored to the specific conditions of individual farms, making it difficult for farmers to identify the right time to intervene. By combining stress detection and treatment in a single platform, our technology reduces unnecessary chemical inputs, minimizes waste and runoff, lowers daily labor requirements, promotes faster plant recovery, and helps maintain more stable crop yields during challenging growing seasons.

Our Solution

Unlike traditional agricultural inputs that rely on centralized chemical manufacturing and reactive spraying, our system introduces novel solutions. Instead of shipping chemical bottles to the field, we deploy an on-site,

on-demand field factory directly to the farm. This bridges the gap left by standard solutions, which either rely on chemical inputs that cause runoff, basic biostimulants that lack automatic delivery, or passive alert sensors that identify stress but rely on farmers to dilute and mix protectants on their own. Our bioreactor uses engineered *Bacillus subtilis* to continuously evaluate crop conditions by pairing real-time field sensors with predictive stress forecasting. When environmental or biological threats pass a safety threshold, the device automatically synthesizes targeted protectants to address climate stress and plant disease precisely when and where intervention is required. Built with a flexible, modular design, the system can be adapted across different crop types and growing setups. It also takes into great consideration of biosafety, ensuring engineered bacteria remain safely contained inside the hardware to allow only active protectants to reach the crops. From our lab testing, by using our ACCD aftertreatment, the fresh weight of Arabidopsis has doubled in comparison with control treatment testing with TB. We predict that could be cross-applied to other crop types in the future with further testing. Furthermore, our solutions offer a cheaper solution and more competitive price when compared with other companies that offer similar options.

Long-Term Mission and Agricultural Impact

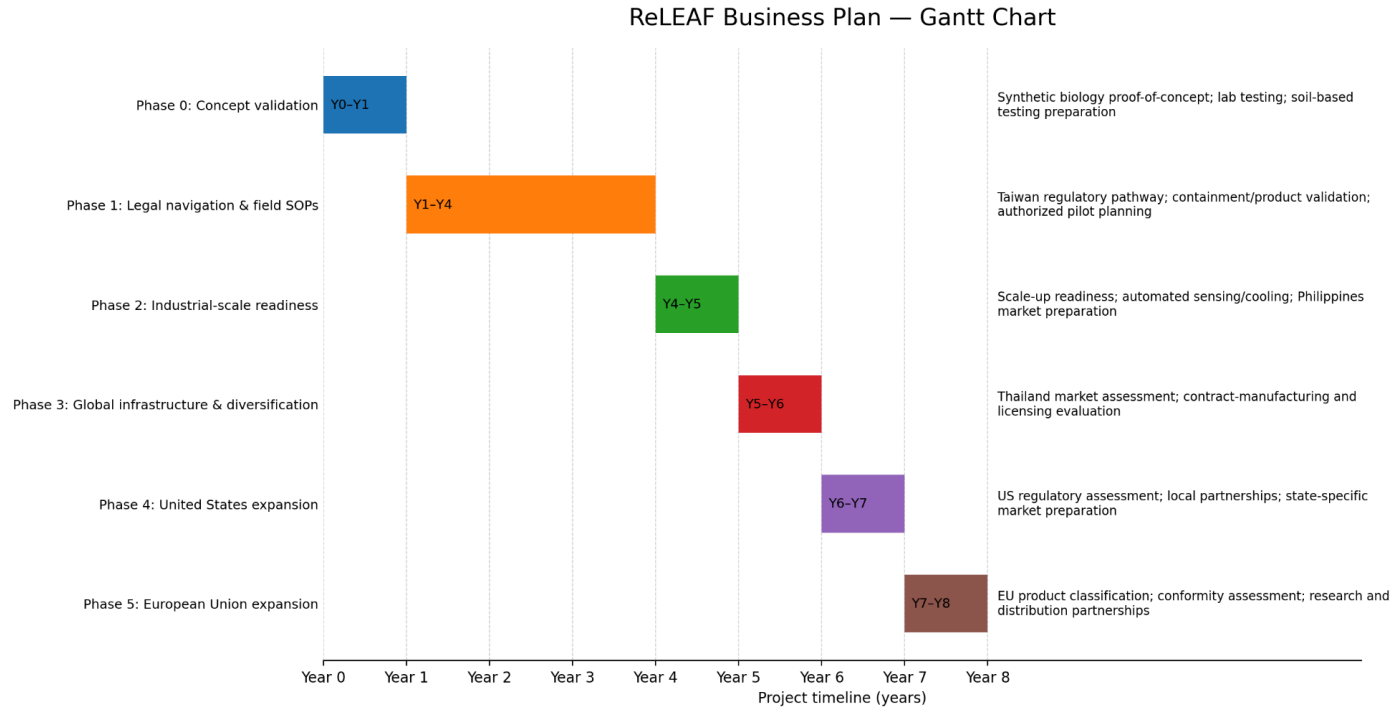
Ultimately, our mission extends to supporting long-term agricultural resilience and vision goals. By collaborating with local agricultural extension stations, research institutes, and cooperatives, we aim to stabilize seasonal farm incomes, prevent climate-driven debt cycles, preserve ancestral land assets, and make sustainable farming viable for future generations. Overall, our system gives farmers an automatic, customizable, more effective, and sustainable way to reduce crop losses from environmental and disease stress while lowering chemical use and improving yield reliability.

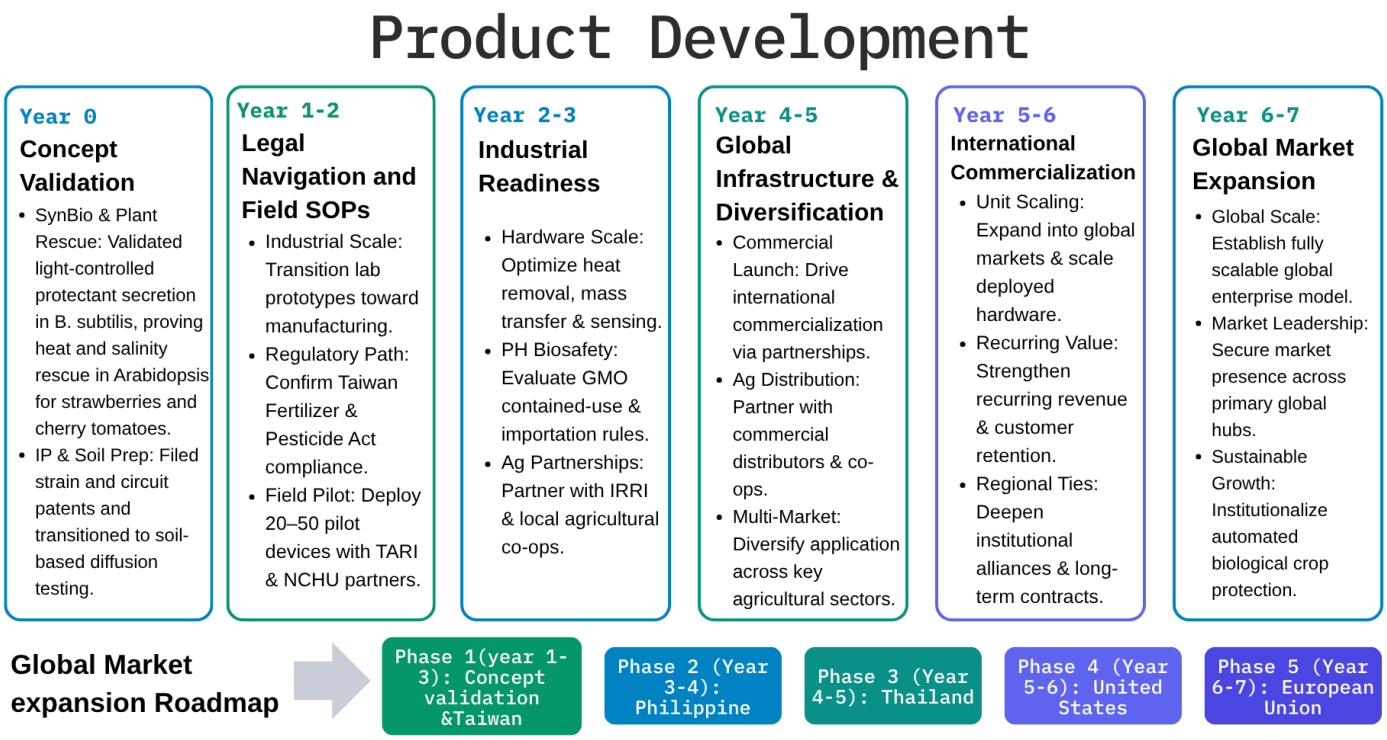
Alignment with United Nations Sustainable Development Goals (SDGs)

Furthermore, this approach directly aligns with several key United Nations Sustainable Development Goals (SDGs):

- **SDG 2 (Zero Hunger):** Protects crop yields against severe climate events and stabilizes farmer incomes without increasing manual field labor.
- **SDG 10 (Reduced Inequality):** Through the social enterprise business model, we allow both small and industrial farmers to have equal access to our technology, regardless of their original purchasing ability
- **SDG 12 (Responsible Consumption and Production):** Minimizes chemical runoff, reduces operational waste, and builds resilient local food production networks.
- **SDG 15 (Life on Land):** Provides a proactive climate-adaptation tool that protects plant health while preserving natural soil microbiomes and land usability

3.4 Product development plan





- **Phase 0: Concept Validation (Year 0)**
    - Our mission is to enhance agricultural resilience against climate-intensified abiotic stressors such as heat waves, salinity and drought by developing an automated, sensor-driven bioreactor. The first

phase focuses on establishing the core synthetic biology framework and achieving laboratory-scale proof-of-concept. We begin by validating the modular pNW33N-L2(sfGFP) shuttle vector and integrated ccaSR light-responsive system in *Bacillus subtilis* to MoClo 2.0 standards. Technical milestones include the optimization of electroporation protocols to achieve high transformation efficiency and the construction of fundamental modules such as the phycocyanobilin (PCB) module, the ccaSm10 light sensor (LSM), and the ccaR transcriptional output module (TOM). Performance is benchmarked by measuring the expression of Late Embryogenesis Abundant (LEA 14) and ACC deaminase protectants under varied light intensities (535 nm green vs. 670 nm red) to determine fold-change efficiency. Concurrently, prototype systems are tested on *Arabidopsis thaliana* models under controlled salinity stress, heat stress, CH peptide, trehalose, ACCD, and LEA by measuring plant fresh weight, chlorophyll content, germination rates, and rescue efficacy against control groups. By late Year 1 around the iGEM Jamboree, the team will demonstrate a fully integrated Machine-Microbe Interface (MMI), validating reversible, light-controlled secretion of LEA 14, ACC deaminase, and BoPep4 within a contained bioreactor system. Strategic intellectual property filings on these optimized microbial strains and responsive circuits will establish a foundation for future industrial engagement with partners. Prototypes are tested on *Arabidopsis thaliana* models under controlled salinity stress. Crucially, year 1 research includes developing the scientific theory and data models to link these lab results to the specific vulnerabilities and economic importance of our target crops including strawberries and tomatoes, like cherry tomatoes. As phase one concludes, our focus shifts from concept validation to real-world soil integration, such as soil testing in the lab, by moving from liquid media to a soil-based lab environment to characterize protectant diffusion and efficacy.

- **Phase 1: Legal Navigation and Field SOPs (Year 1 to 4)**
    - During this phase, ReLEAF will prioritize understanding and complying with Taiwan's regulatory framework before beginning agricultural pilot programs or commercial sales. The first step is to consult Taiwan's Ministry of Agriculture and the relevant authorities to determine how the final cell-free protein and peptide formulation should be classified under the Fertilizer Management Act. Although the existing fertilizer category of Plant Growth Auxiliary Materials may be relevant, ReLEAF will confirm whether its specific formulation and intended claims qualify before pursuing fertilizer registration. If the product's intended function or marketing claims involve pest or disease control, the Pesticide Management Act may also apply, requiring further classification review with the appropriate pesticide regulatory authority. ReLEAF will additionally examine the Organic Agriculture Promotion Act and relevant organic certification requirements to determine whether the product can be used in certified organic farming systems. Regulatory assessment will also address the engineered Bacillus subtilis culture used inside the bioreactor, separately from the final cell-free product. ReLEAF will consult the relevant Taiwanese authorities to clarify requirements for laboratory research, containing production, storage, transportation, and waste disposal involving the engineered microorganism. The 0.22 μm hollow-fiber membrane is designed to retain bacterial cells, but its containment performance must be experimentally validated. Testing should confirm the absence of viable production organisms in the final output and assess residual recombinant DNA using documented methods and detection limits. The bioreactor itself will undergo a separate assessment of applicable equipment safety requirements, including electrical safety, pressure-related risks where

relevant, leakage prevention, and cleaning procedures. During this phase, ReLEAF will document the applicable registration pathway, testing requirements, labeling obligations, and evidence needed to demonstrate product safety and efficacy in Taiwan. Consultation with potential research partners, such as the Taiwan Agricultural Research Institute, Tainan District Agricultural Research and Extension Station, National Chung Hsing University, and Academia Sinica, may support product validation and trial design. Agricultural pilots and commercial deployment will proceed only after the relevant regulatory requirements and necessary permissions have been confirmed. This approach will allow ReLEAF to establish a Taiwan-specific compliance roadmap before expanding its operations. ReLEAF's international expansion strategy is designed to balance agricultural market potential with regulatory feasibility, geographic accessibility, and the practical needs of its target customers. Taiwan will serve as the initial development and validation market, allowing us to establish technical performance, clarify regulatory requirements, and refine its business model. The Philippines will be the first international expansion market because of its strong alignment with smallholder agriculture and its potential for cooperative-based deployment. Thailand will follow as ReLEAF builds regional operating experience and develops more scalable manufacturing and distribution arrangements. The United States and European Union will be evaluated at later stages because entering these markets may require greater regulatory preparation, local partnerships, and investment in market-specific compliance. This sequence allows ReLEAF to build evidence and operational capacity progressively while keeping each expansion conditional on product validation, regulatory requirements, available resources, and demonstrated market demand. For the break-even point, we estimated deployment packages for one hectare farm. This is mainly because 99.2% of Taiwan's farmland is about 1.1 hectares (*An Overview of New Agricultural Policies for Taiwan in 2018*, 2018). For ReLEAF, the break-even analysis estimated how many deployment packages the business must sell to recover its total fixed cost of US$334,053 over the first three years. The calculation divides the total fixed costs by the estimated profit from each deployment package depending on whether the farm grows small-fruit or large-fruit tomatoes as well as crop types. Therefore, to cover up $627,000 in fixed costs over the first four years, we need to sell about 230 hectares' worth of product, bringing in roughly $2,800 in profit per hectare. We plan to price each package at $5,500 in year 1 which is double our production cost and charges around $90 a year for refills. We will break even in year 2, need up to $196,000 in peak funding, and hit a pre-tax IRR of 24%, meaning that we will be able to give investors 2.3times their money back by year 5. This indicates that we will be able to build the financial foundation needed to expand our market globally.

- **Phase 2 Industrial-Scale Readiness and Global Market Integration (Year 4 to 5):**
    - After successfully integrating our product into Taiwan's local market, phase 2 marks the transition from local to industrial scale production and entry into the global agricultural market. We will address the challenges of mass transfer, heat removal, and oxygen distribution by incorporating automated cooling and advanced sensing to prevent the accumulation of inhibitory metabolic byproducts. We will also develop a product-development pipeline by starting with computational modeling, followed by the development of stress-tolerant protectants, and concluding with field validation under relevant agricultural conditions. This phase will continue into Phase 3. Some major shifts will include moving from direct sales toward licensing and evaluating the potential for environmental data services as a

future revenue stream. The data model will collect real-time environmental and operational data, which can be used internally to improve the business by adjusting and optimizing products and services. External data licensing may be explored in the future, subject to data quality, ownership, privacy, and market-demand assessments. In the second half of Year 2, we plan on expanding our market into the Philippines as our primary international expansion market. The Philippines presents an opportunity to evaluate ReLEAF's agricultural applications in a market focused on food security and agricultural sustainability (Monzon, 2025). Before commercialization, we will confirm the applicable regulatory requirements for equipment importation, the contained use of genetically modified microorganisms, and registration of the final cell-free product. The Philippines involves separate regulatory assessments whose applicable requirements and timelines must be confirmed with the relevant authorities. First, we will ensure the bioreactor hardware meets the requirements for equipment importation and applicable distribution and safety standards. The relevant requirements and estimated timeline will be confirmed during regulatory consultation. Second, the engineered culture will be assessed under the applicable Philippine biosafety framework. We will consult the National Committee on Biosafety of the Philippines (NCBP) under the Department of Science and Technology (DOST) to clarify the requirements for research, contained use, transport, risk assessment, and any necessary authorizations (National Committee on Biosafety of the Philippines – DOST, 2023). Third, the final cell-free protectant output will be assessed with the fertilizer and Pesticide Authority (FPA) to determine its appropriate classification and registration requirements. The timeline for confirming Philippine biosafety and product registration requirements subject to the final product classification and regulatory review. This will help ReLEAF establish a clear compliance pathway before beginning any activities that require authorization. We will also seek to partner with the International Rice Research Institute (IRRI), local universities, and agricultural cooperatives to run authorized pilot programs, build regional trust, and validate the agronomic value of our cell-free protectant delivery system (International Rice Research Institute, 2018). Based on pilot results and regulatory permissions, we will develop initial commercial agreements and assess licensing opportunities.

- **Phase 3 Global Infrastructure and Multi-Market Diversification (Year 5 to 6):**
    - By Year 4, the project transitions into further international commercialization, expanding its operational footprint from the foundation markets of Taiwan and the Philippines into Thailand. This phase focuses on establishing early commercial scale as well as multi-market income. To support deployment across multiple international territories, the manufacturing part shifts from localized assembly toward globalized contract manufacturing where it is commercially and operationally feasible. Expanding internationally requires moving beyond a one-size-fits-all product strategy to address highly varied regional weather patterns and environmental stressors. During this phase, we will expand further into Southeast Asia, specifically in Thailand, adapting our product and service model to local agricultural needs and regulatory requirements. Market entry into Thailand involves several regulatory assessments, with the timeline to be confirmed through consultation with the relevant authorities. First, the bioreactor hardware will be assessed under applicable Thai industrial standards and equipment import requirements to verify conformity and determine the requirements

for commercial distribution. Second, the engineered culture will be assessed under Thailand's applicable biosafety framework for genetically modified microorganisms. We will consult relevant biosafety authorities and refer to applicable guidance, including BIOTEC's biosafety guidelines where relevant, to clarify requirements for risk classification, physical containment, research and industrial-scale use, waste treatment, transport, monitoring, and emergency procedures. Third, the cell-free protectant output will be assessed to determine its appropriate classification under the Thailand Fertilizer Act B.E. 2518 (1975) or other applicable regulations. ReLEAF will consult the Ministry of Agriculture and Cooperatives and relevant authorities to confirm product classification, composition and quality requirements, testing procedures, and registration documentation before pursuing commercial distribution. To support market entry, ReLEAF will seek partnerships with Thai agricultural research institutions, universities, distributors, and agricultural cooperatives. These partnerships will help evaluate the product's performance under local environmental conditions, identify suitable crops and farming systems, and support authorized pilot programs. Commercial expansion will depend on successful product validation, confirmed regulatory requirements, and demonstrated local demand. To support international growth, ReLEAF will also evaluate separating the bioreactor hardware service from the biological product supply where this is commercially appropriate and legally permissible. This approach may provide greater flexibility in equipment distribution, licensing, and local partnerships while maintaining separate oversight of the engineered production culture and final cell-free product.

- **Phase 4 (Year 6 to 7):**
    - By Year 5, ReLEAF will expand its international operations into the United States, building on the technical validation and market experience gained in Taiwan, the Philippines, and Thailand. The United States offers a substantial agricultural market, with opportunities to work with commercial farms, agricultural research institutions, and premium-crop producers. However, its geographic distance from Taiwan and decentralized regulatory system make market entry more resource-intensive. During this phase, ReLEAF will assess commercial demand, identify suitable agricultural sectors, and establish partnerships to support research, manufacturing, distribution, and technical services. Before commercialization, ReLEAF will evaluate the applicable regulatory requirements for the bioreactor, engineered *Bacillus subtilis* production culture, and final cell-free protectant separately. For the engineered production organism, the Environmental Protection Agency's (EPA) biotechnology framework under the Toxic Substances Control Act (TSCA), particularly 40 C.F.R. Part 725, will be a principal pathway to investigate. ReLEAF will determine whether its specific strain and proposed manufacturing activities qualify for an exemption or require notification and review. Any applicable requirements for research, contained production, transport, and waste management will also be assessed. In parallel, ReLEAF will investigate whether the United States Department of Agriculture's Animal and Plant Health Inspection Service (USDA-APHIS) has jurisdiction over any aspect of the organism or its use, depending on the organism's characteristics and the applicable regulatory provisions. The final cell-free protectant will undergo a separate classification assessment. ReLEAF intends to position its product for abiotic-stress resilience rather than pest or pathogen control. However, its final classification will depend on the formulation, intended function, and marketing claims. The EPA's requirements under the Federal Insecticide,

Fungicide, and Rodenticide Act (FIFRA) must be considered if the product's claims or function fall within the pesticide or plant-regulator definitions. Fertilizer and plant-biostimulant requirements may also apply at the state level, meaning that ReLEAF will need to assess the requirements of each target state rather than assume that one federal registration pathway applies nationwide. The intended use of ACC deaminase (ACCD), which is being investigated for heat-stress mitigation through its effects on stress-related ethylene, will receive particular attention during classification because plant-growth-regulator interpretations may depend on the specific claims and regulatory context. The bioreactor will be assessed separately for applicable electrical, mechanical, pressure-related, installation, and equipment-safety requirements based on its final commercial design. ReLEAF will also validate the cell-free output by testing for viable production organisms and assessing residual recombinant DNA using documented methods and detection limits. The 0.22 μm hollow-fiber membrane will be described as designed to retain cells until its retention performance has been experimentally validated. These results, together with product composition, stability, safety, and efficacy data, will support regulatory consultation and commercial preparation. To facilitate market entry, ReLEAF will seek partnerships with agricultural research institutions, commercial farms, local distributors, and manufacturing or technical-support providers. The company will evaluate licensing, local manufacturing, and distribution arrangements to reduce the operational burden of entering a distant market. Product specifications, labeling, and distribution strategies will be adapted to the requirements of the selected states and customer segments. Commercialization will proceed only when the relevant regulatory requirements have been clarified, necessary approvals or registrations have been obtained, and market demand has been demonstrated.

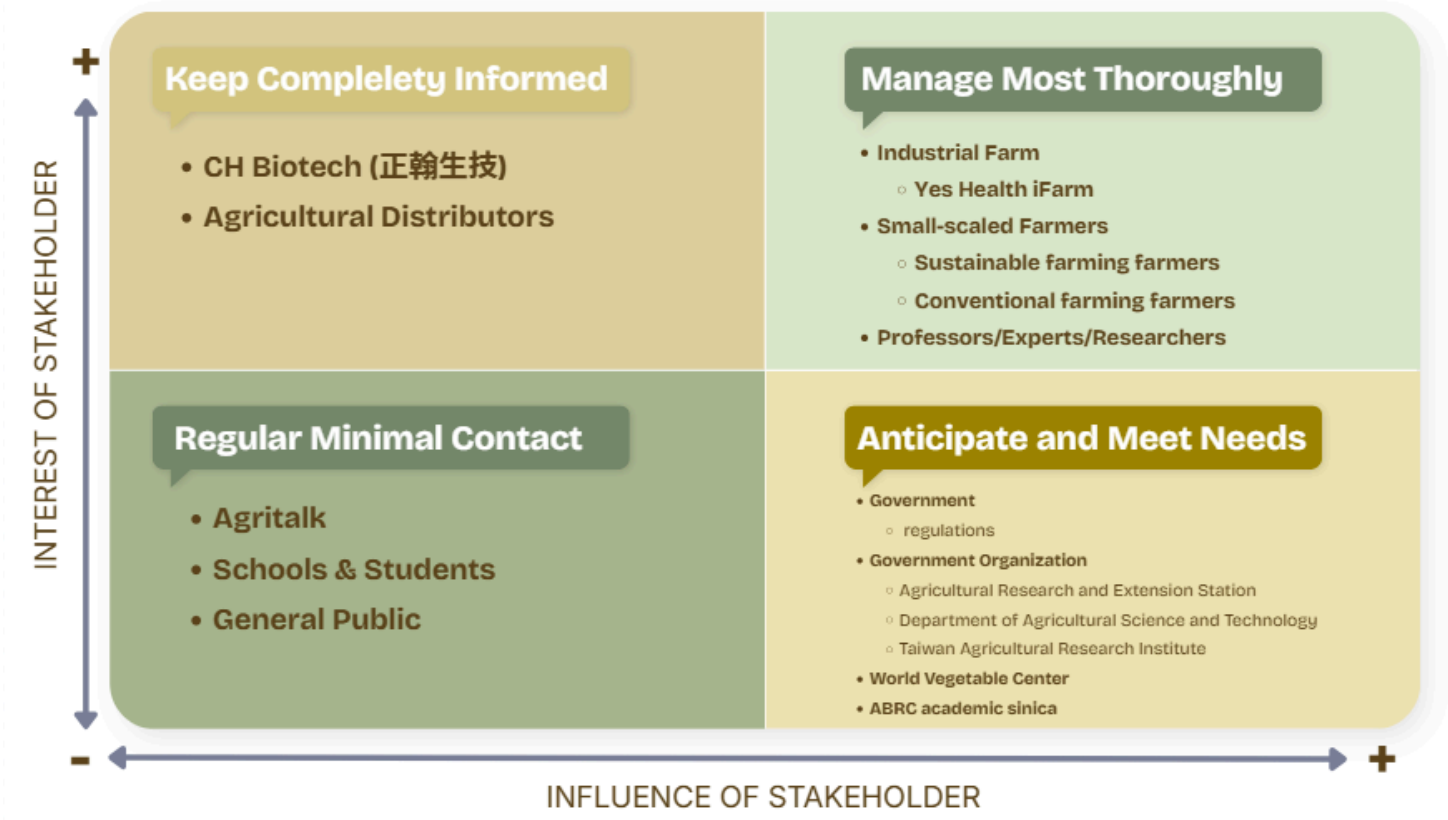
- **Phase 5 (Year 7 to 8):**
    - By Year 6, ReLEAF will investigate expansion into the European Union, building on its experience in earlier markets and evaluating opportunities across selected member states. The EU offers a significant agricultural market and a shared regulatory framework for certain fertilizing products. However, market entry will depend on the composition of ReLEAF's final product, its intended use, and its compliance with relevant regulations. During this phase, ReLEAF will identify suitable member states, target crop sectors, potential research partners, and distribution channels while evaluating the commercial feasibility of introducing its cell-free protectant platform. One potential regulatory pathway for ReLEAF's final cell-free product is Regulation (EU) 2019/1009, which sets out requirements for EU fertilizing products. Under this regulation, Product Function Category (PFC) 6 covers plant biostimulants, including PFC 6(B), which applies to non-microbial plant biostimulants. Since plant biostimulants may be used to improve plants' tolerance to abiotic stress, this category could be relevant to ReLEAF's intended applications. For example, ReLEAF is investigating BoPep4 for salinity tolerance, AtLEA14 for drought and osmotic-stress protection, and ACC deaminase for heat-stress mitigation. However, a product's intended function does not automatically make it eligible for this category. ReLEAF will need to confirm whether its final protein- and peptide-based formulation meets the applicable PFC requirements and whether its ingredients qualify under the relevant Component Material Categories (CMCs). The product must also satisfy the applicable requirements for composition, safety, efficacy, labeling, and conformity assessment before it can be marketed through the EU harmonized framework. The engineered *Bacillus subtilis* culture used to produce the protectants will be assessed separately from the final cell-free product. ReLEAF will

review the EU's contained-use framework, including Directive 2009/41/EC where applicable, to determine which requirements apply to its production system and planned activities. This assessment will cover the handling, storage, transport, containment, and disposal of biological materials and waste. Even if the final cell-free product qualifies for the PFC 6(B) pathway, this would not automatically authorize the use or distribution of the engineered production organism. ReLEAF must therefore establish the regulatory feasibility of its manufacturing process independently of the final product's classification. The bioreactor will also require a separate assessment based on its final design and intended operating conditions. Relevant requirements may include machinery and electrical safety, as well as pressure-equipment rules where applicable. For example, Directive 2014/68/EU may apply if the equipment falls within its scope. ReLEAF will identify the relevant equipment standards and conformity obligations before finalizing the commercial design and preparing the required technical documentation. To support its entry into the EU market, ReLEAF will seek partnerships with European agricultural research institutions, universities, distributors, and potential commercial users. These partnerships could help evaluate product performance under local agricultural conditions, adapt the platform to regional needs, and establish distribution and technical-support networks. ReLEAF will also review its product specifications and labeling against EU requirements. Data collected from earlier markets may help support this process, but its relevance and acceptability for EU regulatory purposes must be assessed rather than assumed. ReLEAF will proceed with commercialization only after clarifying the product's regulatory classification, confirming that its formulation meets the relevant material eligibility requirements, completing the required conformity assessment, and establishing sufficient market demand. If the harmonized plant-biostimulant pathway is not suitable for the final formulation, ReLEAF will investigate whether another lawful route to market is available before committing to commercial expansion.

4. Market Analysis

4.1 Stakeholder Analysis

Stakeholder Map



The diagram is an Influence–Interest Grid, which categorizes stakeholders based on their level of influence over the project (power) and their engagement or concern with the project's outcomes (interest). This helps us decide how to allocate attention and resources to each group.

Our social business model canvas is designed for social enterprises that aim to create impact through subsidising farmers' purchases of our product and help less wealthy farmers to afford technology that can improve their farming while sustaining financial success. It recognized that the paying customers, who receive a product that reduces input costs and protects plants from stress, are not the same person who benefits from the mission. Therefore, in our business plan, we aimed to increase the number of farmers reached, reduce pesticide applications per season, and reduce crop loss.

Manage Most Thoroughly

- Small-Scaled Farmer: We directly impact their livelihood, reduce their crop loss, and protect their health to reduce chemical exposure. However, as individual smallholders, they have less influence on the market. Since our business plan aimed to create impact, we should keep them deeply involved and informed through the progress.
- Conventional farming farmers: Farmers who rely primarily on chemical fertilizer and pesticides often prioritize yield and cost. This group represents part of the smallholder farmers we are targeting. According to the Agriculture and Food Agency, Ministry of Agriculture, only 3.47% of Taiwan's total farmland is organic and eco-friendly by 2024 (Agriculture and Food Agency, Ministry of Agriculture, 2025). Globally, 2.1% of agricultural land was managed organically in 2023, which, in other words, means over 97% of agricultural land is not organic, so if ReLEAF can't earn the conventional farmer's trust, the addressable market will shrink (Research Institute of Organic Agriculture FiBL, 2025). If we successfully let the conventional farmers replace their chemical fertilizer with our product, it can prove that ReLEAF delivered a better yield stability and lower cost while being sustainable.
- Sustainable farming farmers: Sustainable farming farmers are the ones who prioritize reduced chemical input and soil health. Convincing this segment can prove that our engineered *Bacillus subtilis* will not cause pollution while being sustainable, and earning their adoption would serve as an additional point of validation and credibility for our project.
- Industrial Farm: Large-scale commercial operations that represent massive market validation and revenue potential. Through industrial farms, we can see what we are missing in our design or business plan clearly. By collaborating with them, we tailored enterprise solutions and collected data on crop growth from them using our product in order to prove yield consistency.
- **Yes** Health iFarm: Yes Health iFarm is a Taiwan commercial indoor vertical farm. Although we positioned them as a competitor rather than a collaborator, they are still a valuable stakeholder to us since they do precision farming to collect data and increase crop yields with LED lighting and pesticide-free growing methods, which is similar to our main goal. Because their model relies on indoor cultivation, which is different from our open-field bioreactor approach, we can also learn from their idea of not using pesticide while not copying their direction.
- Professors / Experts / Researchers: The experts can provide us feedback about the design through technical consultation and can shape our project direction and experimental design. Their high influence can be seen through their expertise and credibility. In return, we can provide data for them to further study their expertise.

Keep Completely Informed

- CH Biotech: Taiwan biotech company specializing in plant growth and systemic biostimulants. We can consult with them about our protectant since they are already selling protectant in Taiwan and other countries, so we can learn from how they expand their business and the regulations in not only Taiwan but globally.
- **Agriculture** distributor: Agriculture distributors understand farmers' demand more since they interact directly with the farmers actively, and they hold practical knowledge of purchasing behavior, price sensitivity, and product concerns that we would otherwise have to gather more slowly. They also have a more complete

supply chain, which will allow us to reach the farmers easily. Agriculture distributors may be more likely to want to sell our product, understand the farmer's demand, and have a more complete supply chain.

Anticipate and Meet Needs

- Government (for regulations): The government doesn't necessarily care about how our business plan benefits the society. However, they care immensely about environmental safety laws and agricultural standards; a single policy shift can make or break our product compliance. To ensure our product fits the regulations, we should ensure all certifications are up to date and comply with safety and environmental laws.
- World Vegetable Center: An international non-profit organization with massive global influence in agricultural development. Our project might just be a small part compared to them, but we should still keep track of them since their support or network could scale our social impact globally.
- Government Organization (Agricultural Research and Extension Station, Department of Agricultural Science and Technology, Taiwan Agricultural Research Institute): Researchers in these departments hold immense scientific credibility. By contacting and keeping track of these departments, they can consult us about real field conditions in Taiwan and how we should design our product to avoid breaking the law and regulations. In return, we can also provide data for them to further study their expertise.
- Academia Sinica: As Taiwan's top academic institution, Academia Sinica leads cutting-edge research across disciplines, including biotechnology and materials science. Their input enhanced our design in different parts, including the gene circuit, engineered *Bacillus subtilis*, and hardware. By collaborating with their researchers, we can learn from their research and gain more feedback on the design.

Regular Minimal Contact

- Agritalk: A company that provides farmers with deep-dive analysis on farm production, agricultural policy, new technology, and commodity markets. They are classified as our competitor but won't dictate our operational success.
- Schools & Students: Our outreach involved hands-on science education with students, introducing synthetic biology and sustainability. Although they won't affect the development of our product, they still represent our commitment to inclusivity and long-term behavior change through education.
- General Public: We monitored public interest through interviews, workshops, and social media. Engaging with the public helped amplify our impact, though they hold limited power over the development.

4.2 Industry Analysis

Overview of the Global Protectant Industry

The global protectant industry is within the agricultural sector, valued at 77.17 billion USD. As rapid climate changes and a growing population demand for more and better crops (Rongkali, 2026), this market is rapidly expanding. The more specific market that we will be in is bioprotectants, or biocontrol agents, where biological parts are used to create protectants instead of the chemicals that usually dominate the market. This market was valued at 9.4B USD in 2024 and is projected to reach 22.8B by 2030, providing strong proof of future growth and ensuring that the market won't fail. In comparison, the chemical fertilizer/protectant market has already reached 216.52 billion USD in 2025 and is projected to reach 299.69 billion USD by 2030 (The Business Research Company, 2026a, 2026b).

Key Companies in Bio-Protectants

Key bioprotectant companies are driving the shift toward more sustainable crop protection by developing biological alternatives to conventional chemical pesticides. Global agricultural leaders such as Bayer AG, BASF SE, Syngenta AG, and Corteva Agriscience have expanded their biological crop protection portfolios through investment in biopesticides, microbial products, and integrated pest management solutions. Specialized biosolutions companies, including Koppert Biological Systems, Valent BioSciences, Andermatt Biocontrol, and Rovensa Next, focus primarily on microbial, botanical, and beneficial insect technologies that reduce pest and disease pressure while supporting environmental sustainability. Together, these companies are accelerating innovation in the rapidly growing bioprotectants market through research, strategic partnerships, and expanded commercial production of biological crop protection products.

In Taiwan, a key company is CH Biotech (正翰), which we have already had contact with. This company focuses primarily on spray-type bioprotectants that are similar to some extent to ours and can be a potential partner in the future. Another key company is Agriweather. Agriweather is a company that produces a sense and regulation system that is very similar to ours, where they install sensors and a general control system that connects to the farmer's irrigation system so that it can take care of things like irrigation, fertilizer application, and pesticide application.

Market Opportunities

As resistance to chemical pesticides continues to increase, farmers are being pushed to seek alternative crop protection methods, creating growing interest in bioprotectants. At the same time, the global climate movement has accelerated the adoption of more sustainable farming practices. However, many synthetic biology-based solutions have struggled to align with organic farming standards, limiting their ability to fully benefit from this trend. Meanwhile, climate change is making weather patterns increasingly unpredictable, introducing new environmental stresses that traditional farming experience alone can no longer reliably anticipate or manage. This creates a need for intelligent, automated systems that can respond to changing conditions in real time. Our bioreactor addresses this gap by continuously sensing environmental and plant stress signals, producing only the specific bioprotectants necessary. Additionally, because our bioreactor has a sensing, response, and production system all in one, our product also has competitive pricing in comparison to other similar products, which can range from 300 to 2,500 USD for just sensing or just protecting production. Unlike conventional chemical protectants, which are applied on fixed schedules regardless of actual crop conditions, our system offers a responsive and adaptive approach that reduces unnecessary inputs while simplifying farm management. By combining automation, precision, and sustainable

bioproduction, our bioreactor represents a new generation of agricultural technology designed to help revitalize farming in an increasingly uncertain climate.

Challenges

Despite all these opportunities, our product will undeniably face several challenges on the market. Before our product can even hit the market, there are things that might stop it in its tracks, more specifically government regulations. Due to our product including GMOs, it will face legal hurdles such as extensive testing, heavy usage restrictions, and close government surveillance, as GMOs are treated with much caution by several countries and key markets, such as the European Union, Australia, and our home country, Taiwan. This could also restrict how much of our product we can distribute throughout countries and in Taiwan and may make it so that the person buying our product needs a license to use it. Aside from existing legal challenges, there's also existing stigma regarding synthetic organisms that may push farmers away from purchasing our product, as they fear it could negatively affect the environment or escape the containment of our bioreactor. The *B. subtilis* itself could also present a challenge in that its survival rate may be too low to truly be sustainable, causing a need for frequent switches and creating a high cost to using our product that will likely alienate our target demographic, who are smaller, sustainable farmers.

Strategic Positioning

Our product is the first on the market that is able to respond in real time to the stresses faced by plants, where the protectants are only produced when stress reaches a threshold level and at the exact amount that is suitable for the farm it is used on. This type of customisability and specificity is unique to our product, and unlike current solutions, which require farmers to constantly have to interpret when to use protectants, how much to use, and where to use them, risking overuse or underuse that not only harms plants but also the environment. Our bioreactor and *B. subtilis* are also capable of responding to several specific types of stress, instead of just using one cure-all that could be unsuitable to fully tackle the biological effects of a certain stress type on various crops.

4.3 Competitor Analysis

Comparative Table



Our System

Our system: autonomous protection production, sensor-triggered, dual-module, cell-free. No existing product in this quadrant.

Direct Competitors:

Category 1: Sensing and Controlling

Name	Description	Pros	Cons
Agriweather	An AIoT platform pairs soil sensors (mainly drought sensors) with weather news to predict crop growth and stress via an easy mobile app. It calculates precise field needs and controls custom multi-pathway irrigation devices, backed by government agency partnerships.	- Intuitive, accessible mobile app paired with predictive AI stress modeling to optimize water and nutrient dosing. - Strong government agency backing and field-tested credibility.	- High upfront setup cost (USD $1,800 sensor set + USD $2,500– USD $4,600 secretion device). - Acts as an automated valve system without biological synthesis capabilities. - Bioprotectant isn't specifically targeted at any specific stresses. - Uses any fertilizer/protectant the farmer prefers, which can be chemical and isn't completely eco-friendly
AgriTalk Advanced AIoT system	An advanced AIoT (Artificial Intelligence + Internet of Things) precision farming platform that integrates local soil and weather sensors with cloud computing to predict crop diseases, manage pests, and remotely regulate automated fertilization and drip irrigation systems.	- Detects environmental and soil shifts and pushes wireless signals to actuators, instantly adjusting farm conditions. - automatically flags faulty sensors and recalibrates them remotely without requiring constant battery replacements.	- Relies completely on installing and maintaining physically connected physical modules (AgriCtls control boards, sensors, and valves) across the field to function. -Requires initial hardware positioning, calibration, and customized rules mapping. Makes it hard for farmers to use
PlantDitech PlantArray	An automated system that measures how plants grow and use water in real time. It uses sensors to track plant water balance and growth under different conditions and releases water and fertigation according to results automatically.	- Automatic & easy to collect detailed data - Accurate plant growth measurements - Can test multiple conditions at once	- Needs special training - Complex setup - Only for abiotic stress and only releases water and fertilizers dissolved in water

Plant DiTech SPAC Analysts	Cloud-based software that automatically retrieves and processes real-time data from plant sensors and environments to perform statistical analysis, growth modeling, and productivity predictions	- Instantly processes massive amounts of sensor data without manual work. -Tracks complex traits like water efficiency and stress responses	- Requires another specific hardware device from the company to actually function. - Only senses the plants' conditions with no response
Averra Nova Device	A portable, cost-effective diagnostic device that integrates hydrogel-embedded microbial biosensors and advanced microfluidics to provide on-site, real-time testing of bioavailable heavy metals (such as cadmium and manganese) in soil and crops.	- provides accurate field readings in minutes. - Uses advanced microbial biosensing to specifically isolate and measure bioavailable toxins	- relies on physical on-site deployment and proper upkeep of microfluidic channels, and handling delicate hydrogel components.

Category 2: Bioreactor

Name	Description	Pros	Cons
Bio-King	An AIoT-enabled atmospheric fermentation machine that uses specialized microbial packets to brew live bacterial solutions on-farm via a 48-hour incubation cycle. Available in small and large models, it targets tech-forward younger farmers and small-scale operations.	- Validated, market-ready physical hardware deployed across various farm scales. - Reduces long-term input costs by producing treatments locally on the farm.	- A 48-hour incubation lag prevents an immediate response to sudden environmental stress. - Relies on living bacteria and lacks integrated sensors to self-trigger synthesis.

Category 3: Bioprotectant

Name	Description	Pros	Cons
正瀚生技 **(CH Biotech)** 富肽 (Radiant)	An advanced agricultural signaling peptide formulation designed to regulate core plant physiology, enhancing photosynthesis, reducing reactive oxygen species (ROS) damage, and preserving cellular membrane integrity during sudden climate stresses like heatwaves, droughts, or frosts.	- Directly addresses modern climate volatility by protecting yields and maintaining strong metabolic functions during acute weather-induced plant stress. - concentrated and chemically optimized to match eco-friendly and organic standards	- No Real-Time Detection
APOLO Biotech Exogenous dsRNA Platform (Pest Gene Silencing)	An advanced biotechnology platform utilizing topically applied, non-GMO double-stranded RNA (dsRNA) molecules to trigger RNA interference (RNAi), effectively silencing essential genes in specific fungal pathogens like *Botrytis* and *Fusarium* to act as a precise "vaccine" for plants.	- Addresses both abiotic and biotic stress with more focus on biotic stress -Only affects specific fungal genes - Eco-friendly alternative to fungicides - Customizable	- no real-time detection - The protective vaccine effect relies entirely on human precision during spraying - Not yet on the market but developing its first products
APOLO Biotech Plant Modulation RNA Platform	Topical, non-GMO biological spray that uses exogenous noncoding RNA molecules to temporarily modify a plant's internal epigenetic markers, helping crops adapt to severe abiotic environmental changes like sudden frost or heatwaves without changing the plant's DNA.	- Directly optimizes the plant's natural hormone homeostasis on demand, protecting vital tissues from catastrophic climate stress. - Degrades naturally in the environment without creating synthetic chemical buildup in the food or soil	- Manual Mixing & Spraying: They must be manually mixed and physically misted onto fields - The product does not contain built-in digital intelligence or IoT sensors

Indirect Competitors:

Category 1: Chemical Fertilizers

Name	Description	Pros	Cons

Pinsoa Chemical Industry Plant Hormones	Artificially synthesized and naturally derived plant growth regulators (PGRs) designed to control and optimize crop development stages, including rooting, cell division, growth retardation, and yield improvement.	- Allows growers to trigger specific traits, such as breaking seed dormancy or accelerating rooting. - Offers a large catalog of specialized chemical formulas tailored for different crop types.	- Users must manually compute, dilute, and mix precise chemical ratios. - There is no built-in tracking, data sensing, or automated feedback loops.
威玖生技 **(Vigor Agrochem)** 旺春風 (Wangchunfeng)	A specialized liquid plant physiological stimulant designed for foliar application during flowering and early fruiting stages to enhance crop resilience against weather stresses, promote flower preservation, and stimulate initial fruit growth.	- Effectively stimulates physiological activity precisely when plants are vulnerable, helping to minimize flower/fruit drop during weather shifts. - easy to apply, allows crops to develop root systems rapidly	- It offers no data sensing, feedback loops, or real-time physiological stress indicators. - Incorrect manually calculated dosage or wrong weather application window can hurt plants or fail to save the crop due to human application error.
Acadian Stella Maris®	A premium liquid sea kelp concentrate derived from 100% *Ascophyllum nodosum* using a proprietary extraction process to retain beneficial bioactive compounds such as alginic acid, mannitol, and oligosaccharides that stimulate root growth and plant resilience.	- Actively triggers antioxidant production and cell water retention to mitigate damage from extreme drought, salinity, heat, and cold - Contains natural chelators that bind to micronutrients in the soil, increasing their availability and transport within the plant.	- must manually measure, dilute, and apply the product via traditional spray tanks or irrigation setups - Offers no integrated data sensing or feedback mechanisms. - relies on human timing

Category 2: Chemical Pesticides/Insecticides

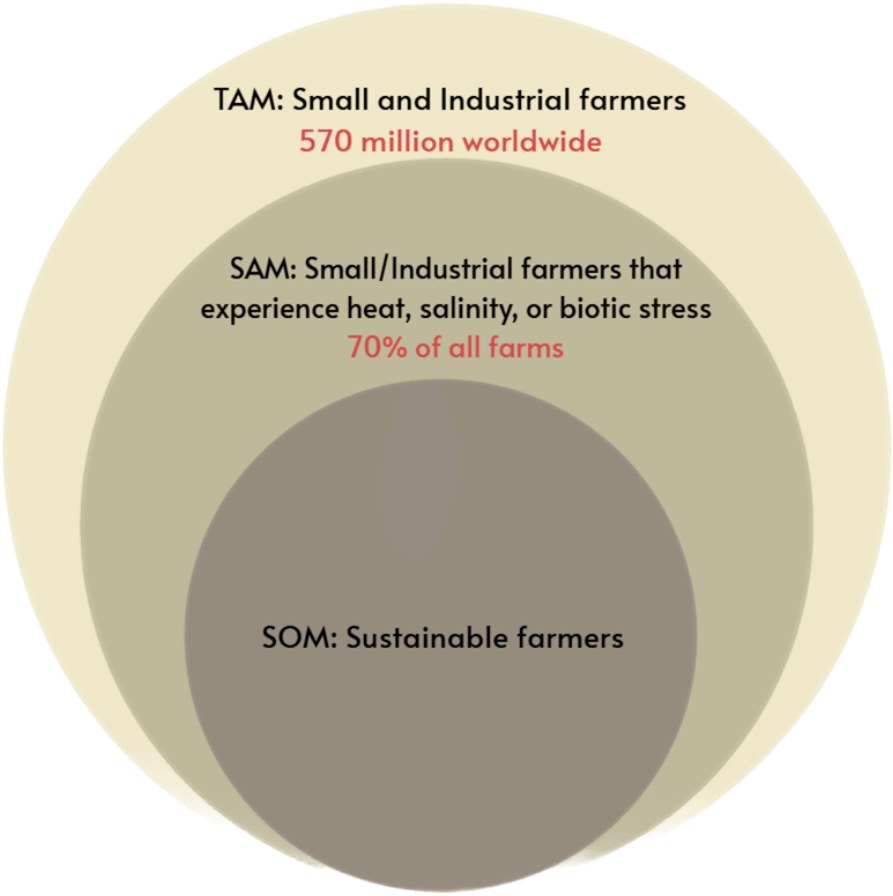
Name	Description	Pros	Cons
Syngenta Karate®	A broad-spectrum synthetic pyrethroid insecticide (active ingredient: lambda-cyhalothrin) applied to control chewing, biting, and sucking pests across major field crops.	- Immediate chemical knockdown against established insect infestations. - Highly stable, low-cost liquid input with long shelf life.	- Exposes crops/soil to persistent chemical residues. - Lacks real-time field sensing, risks off-target toxicity, and promotes chemical resistance over time.
Bayer Crop Science: Bulldock®	A synthetic pyrethroid insecticide (active ingredient: beta-cyfluthrin) formulated for rapid control of agricultural pests on fruit, vegetable, and grain crops.	- Fast-acting contact and ingestion action to kill active pests quickly. - Well-established commercial distribution network. - Cheap	- Entirely dependent on human application timing with no automated threat detection. - Chemical-based mechanisms cause broad environmental toxicity and offer no abiotic stress protection.

Summary

Our project stands out from competitors by establishing a decentralized, on-site, and on-demand automated field-deployed factory. Unlike hardware-only systems like AgriTalk or Smart Control Boxes that provide data or automation without any physical treatment, or tools like Averra that detect heavy metals but cannot fix them, our system directly addresses current solution gaps by producing and applying targeted treatments right where they are needed. This eliminates the labor-intensive, completely manual spraying required by standard biostimulant and biopesticide platforms like Robigo, APOLO Biotech, BioWorks, or Flourish, minimizing human error and optimizing application timing. This enables high ecological efficiency and precise crop protection with minimal waste, offering immense scalability and operational cost reduction over time.

By generating solutions on-demand right on the farm, our system offers targeted activity, enabling a cleaner, safer, and more autonomous farming process with lower resource inputs and broader crop compatibility.

4.4 Target Market Research



Total Addressable Market: About 570 million farms worldwide (Ritch and Roser, 2022), with over 916 million people working on farms and the agricultural sector in 2023 (FAO, 2025).
Serviceable Available Market: 399 million farms, or 70% of farmers, report being affected by climate stress (Gutmann, 2023).
Serviceable Obtainable Market: Hard to quantify because it's not something that has a clear definition or method of registration, but there are around 4.7 million organic farms worldwide.

Customers	Customer segmentation
Governments	- Buy from us to distribute to farmers - use in initiatives for precision/smart farming

Small Farmers	- Small farmers that are suffering from heat, salinity, or biotic stress - Subsidized farmers post-globalization - Farmers that lack human labor Example demographic: - Top 100 Young Farmers in Taiwan ○ More likely to adopt new technology and be more open to our tech - Farmers that lack labor ○ Approximately 6% of major farming households experienced labor shortages, with 84% of these shortages attributed to a lack of temporary workers in 2023 (Wang, 2026) - Natural farming farmers ○ Emphasizes not providing external inputs to help plants grow; may be more interested if ReLEAF demonstrates ecological safety - Organic farming farmers ○ Representing 3.47% of the country's farmland in 2024 (Agriculture and Food Agency, Ministry of Agriculture, 2025) - Conventional farming farmers ○ Uses fertilizers and pesticides; might be more open to technology in the farm if it improves yield stability or reduces input costs - Eco-friendly farming farmers ○ Registered eco-friendly farming area stands at 6,708 hectares (Agriculture and Food Agency, Ministry of Agriculture, 2025)
Research institutions	- Buy our data/use it for experiments - Partner for testing
Commercial/industrial farms	- rich enough to bulk buy and afford our product to test out - stable customers to help sustain the subsidies that are given to small farmers - Sustainable commercial/industrial farms

5. Marketing Strategy

5.1 Position and Marketing Strategy

ReLEAF's international expansion strategy follows a phased, regulatory-gated model: Taiwan will serve as the initial validation market, the Philippines will be ReLEAF's first international market, Thailand will represent the next stage of Southeast Asian expansion, and the United States is a longer-term target. This sequence allows ReLEAF to establish technical credibility, demonstrate commercial feasibility, and

strengthen its regulatory dossier before entering markets with different or potentially more complex requirements as to the business plan.

Our logic is that basing on the fundamental distinction between the bioreactor, engineered *Bacillus subtilis* culture, and the final cell-free output are separate regulatory components. Approval or acceptance of one component does not mean total authorization of the other two. ReLEAF will therefore obtain jurisdiction-specific determination for each component before commercial deployment. The timeline below represents our planned development milestones rather than guaranteed approval dates. Progression between markets will depend on regulatory clearance, demonstrated product safety and efficacy, successful containment validation, and sufficient commercial demand.

Organic Farming and GMO Concerns

Organic certification standards often restrict GMOs and products produced using genetically engineered organisms. Although ReLEAF retains engineered *Bacillus subtilis* within a contained bioreactor and collects cell-free outputs, this does not automatically equate to organic compatibility. Compliant measures vary by jurisdictions, so ReLEAF will initially prioritize conventional agriculture while assessing organic eligibility through country-specific regulatory review. No organic-compatibility claims will be made without confirmation.

Regulatory considerations differ across target markets. The European Union's Regulation (EU) 2018/848 imposes explicit restrictions on GMOs and products produced from or by GMOs, while the United States' USDA National Organic Program excludes genetic engineering methods. Taiwan also restricts GMOs in organic production. These frameworks create potential barriers even when the final product contains no viable engineered bacteria.

ReLEAF will therefore separate biological containment from organic certification. Its hollow-fiber membrane is designed to retain engineered bacteria, but containment alone does not prove that the resulting proteins or peptides meet organic standards. Further assessment may require evidence of bacterial retention, membrane integrity, residual genetic material, and the production process. Until eligibility is confirmed by the relevant authorities or certification bodies, ReLEAF will avoid marketing its outputs as suitable for certified organic farming.

5.2 Regulatory Expansion Timeline

Development phase	Regulatory work	Milestone

Phase 0: iGEM project development, Dec 2025 to Nov 2026	Desk review of the three components across eight markets, with an industry briefing from CH Biotech on the competent authorities in Taiwan, the EU and the US. Lab work stays inside iGEM's safety rules, which limit high school teams to Whitelist organisms, so gray mold is not tested this year and BoPEP4 is claimed only for salinity tolerance.	Laws and Regulations analysis published; claims fixed to abiotic stress
Phase 1: soil testing and safety, Nov 2026 to Oct 2027	Classification consultations with the Agriculture and Food Agency and APHIA. Membrane retention and escape-frequency testing with a government-recognized institution such as the Taiwan Agricultural Research Institute or the Tainan District Agricultural Research and Extension Station.	Written answers on product category and contained-use status
Phase 2: Taiwan pilot, Oct 2027 to Oct 2030	Submit the registration dossier: composition, stability, efficacy, safety. Run pilot trials only under the authorizations those answers require.	Registration decision; pilot data for the dossier
Phase 3: Philippine preparation, Oct 2030 onward	Consult the Fertilizer and Pesticide Authority on classification, which names genetically modified organisms inside its fertilizer system (Fertilizer and Pesticide Authority, 2018), and the DOST Biosafety Committee on importing and using the culture (Department of Science and Technology Biosafety Committee, n.d.). Appoint a licensed local importer and an FPA-accredited researcher.	FPA registration pathway confirmed

Phase 4: international deployment, 2030 onward	Country assessments in the order set on the Laws and Regulations page, each treated separately.	Local authorization before any sale
Phase 5: 2031 onward	Thailand and Indonesia. The United States and European Union come later: the US needs both federal and state clearance (U.S. Environmental Protection Agency, n.d.), and in the EU our engineered *B. subtilis* falls outside the four microbe groups permitted for microbial biostimulants, though the cell-free output may fit the non-microbial category PFC 6(B) (European Parliament and Council of the European Union, 2019).	—

ReLEAF will be pushing to expand to other Southeast Asian markets as well as the United States of America or the European Union market as mentioned in the Future Expansions of Laws and Regulations page.

Regulatory Conclusion & Market Prioritization

Our regulatory evaluation positions Taiwan as ReLEAF's development and validation market, with the Philippines as our first international expansion target. Thailand follows the Philippines, while the United States represents a subsequent opportunity despite potentially greater classification uncertainty and overlapping federal and state requirements.

The EU will be assessed in later stages. Although the EU's Regulation (EU) 2019/1009 provides a potential pathway for non-microbial plant biostimulants under PFC 6(B), this classification does not guarantee eligibility or approval. ReLEAF must establish that its final formulation meets the applicable component material, composition, safety, efficacy, labeling, and conformity requirements. The EU therefore remains a longer-term opportunity rather than our initial international market.

Pricing Strategy

Our pricing strategy is designed around seasonal agricultural cash flow. Throughout Taiwan and Southeast Asia, most smallholder farmers earn their income primarily after harvest cycles. To overcome the high upfront cost of hardware, the system is designed as a low-cost reusable base unit paired with replaceable, seasonal peptide cartridges. Target pricing centers around the abiotic stress present during the growing season. Adoption is most likely if one season of the service costs the same as the existing method and if the device reduces repeat spraying labor and crop loss risk. However, farmers now evaluate solutions based on yield stability, affordability, and reduction of financial risk. A possible solution would be a reusable drive and a replaceable cartridge system that aligns best with farmer spending behavior, which would be a low-cost reusable base device and replaceable seasonal peptide cartridges. Many smallholders operate on thin margins and avoid high-risk investments, which means a renting system can lower the financial barrier and improve accessibility for lower-income farmers.

Customer Segments and Approach

a. Agricultural Orgs.: Emphasize reduced chemical use, lower labor costs, and improved crop yields
b. Agricultural Extension Offices: Focus on scalable deployment and benefits for local farming
c. NGOs and Sustainability Programs: Show environmental impact and sustainable farming practices
d. Research Institutions: Emphasize innovation, field validation, and collaboration opportunities
e. Farmers:
    - Young/Teen Farmers: Tech-savvy and open to sustainable solutions, younger farmers act as early adopters. We connect with them through AgTech expos, youth farming networks, and research organizations, sharing our pilot data and research to build technical trust.
    - Senior & Traditional Farmers: These farmers are risk-averse and value proven results. We reach them through local Farmers' Associations, Production and Marketing Groups, and government research stations (TARI). To lower their barrier to entry, we utilize government agricultural subsidies and conduct hands-on, face-to-face field demonstrations

Success Metrics

a. Reduced chemical and fertilizer inputs
b. Improved plant health under stress
c. Less time spent monitoring crops reported by farmers/users
d. Positive farmer feedback and willingness to continue using the system
e. Convert the pilot program into a long-term licensing deal or supply agreement

Initial Sales Strategy

Our initial sales strategy focuses on partnering with organizations rather than individual farmers, allowing for faster adoption, larger-scale deployment, and greater credibility. By working with established agricultural networks, we can efficiently introduce our technology to multiple farms while building trust through institutional support.

Sales Objective

Year 1 (2026)

- First commercial pilot programs will launch in late 2026 in Taiwan
- Secure pilot partnerships with local agricultural cooperatives, research institutions, or district agricultural extension stations.
- Deploy 0–15 initial hardware units in controlled field settings to validate real-world protectant production and cell-free containment.

Year 2 (2027)

- Convert pilot programs into a multi-year commercial service or supply agreement.
- Expand pilot deployments to 20–30 active units across target high-value crop regions
- Achieve initial recurring revenue through consumable media and protectant cartridge replacements.
- Establish a stable market in Taiwan

Year 3 (2028)

- Enter the regional global market in places such as the Philippines.
- Additional agricultural cooperatives and commercial farms, expanding hardware deployments to 150+ units.
- Achieve break-even on regional hardware manufacturing and distribution operations.

Year 4 (2029)

- Expand distribution networks beyond local markets into Southeast Asian markets
- Reach 500+ deployed bioreactor units with a 85%+ retention rate on recurring annual cartridge subscriptions.

Year 5 (2030)

- Scale operations to 500-1000+ active deployments across places in Asia, the US, and Europe.
- Transition recurring cartridge and media sales into the primary revenue driver
- Gaining more attention to generate more income

Sales Channels (All Years)

a. Partner with NGOs and sustainable agriculture organizations to deploy pilot programs
b. Partnering with farmer associations to promote the bioreactor as a communal resource that they all share makes introducing the small-scale bioreactor easier and more approachable
c. Collaborate with research farms and universities for product validation and demonstration and promotion
d. License the automated bioreactor technology to large agricultural organizations or distributors

Sales Process (All Years)

a. Connect with customers through agricultural expos, farming associations, and sustainability events.
b. Pilot Programs: Deploy 5–10 bioreactor systems for one growing season with training and technical support provided through the program
c. Conversion: Turn successful pilots into long-term purchasing or partnership agreements
d. Retention: Provide continuous technical support, system maintenance, customization, and performance monitoring

5.2 Distribution

At the beginning of our product launch, our platform is not big enough to successfully perform a sufficient amount of direct sales, so our distribution will primarily be done through partnering with distributors to sell our product to farmers. This way, we can influence the Taiwanese market without having to expend the time and energy to directly make connections just to sell our product.
Our distribution will primarily be done through selling our product to middlemen and selling internationally, as the laws and regulations in Taiwan may make it difficult for our product to be approved and widely distributed here. Therefore, the best option is to first expand to more open international markets such as the United States of America, the Philippines, Indonesia, and Vietnam. These locations have looser laws on GMOs and AgTech than areas such as the EU or Japan and are highly affected by increasing climate stress, making them markets that should not have a not-insignificant need for our product. Our product will be distributed B2B, indirectly through middlemen that will buy our product and then distribute it afterwards to farmers, as they will have more sway and connections to various types of farmers and areas than us.

5.3 Promotion and Advertising

Our target market is conventional smallholder farmers, and our promotion strategy is built around them. To reach these target farmers where they appear most, we will focus our efforts through local agricultural institutions, extension stations, and cooperatives. To build trust within these farming communities, we use a zero-configuration physical interface designed specifically for them. It eliminates all complex inputs and uses a heavy-duty, single-button system with mechanically keyed cartridges to prove the device requires no digital skills for our target farmers to operate. Furthermore, our precision-farming bioreactor is especially appealing

to farmers because of how our system is automated and will not require extra work from farmers.

To convince these smallholders to try the system and lower their financial risk, we will offer them a free trial period. We will also use a seasonal rental or shared ownership model. This keeps the cost equal to or less than what they already spend on chemical fungicides, which lowers the financial barrier for these farmers. To establish early trust, we introduce the hardware first to show operational reliability before introducing the cartridges. Once these target farmers see a 10% to 15% increase in crop yields and a 20% to 30% drop in chemical spraying, we will launch a referral discount program. This rewards our target farmers for recommending the system to others, using their networks to speed up adoption. Additionally, we can also work with the Taiwan Agricultural Research Institute to promote and push our product to farmers, as they often subsidize or pay for agri-tech products for farmers in the initial months of usage, allowing farmers to decide whether they want to continue using the product before they commit to a purchase.

5.4 Sales Strategy

Marketing Strategy Timeline



Our sales strategy will launch in 2027, aiming to drive the adoption of our automated bioreactor system across key agricultural sectors, including local farming cooperatives, the general public, and regional extension networks with a strong interest in climate resilience and sustainable crop protection. This strategy is designed to build credibility and trust and expand our market reach while offering financial options that can cover both immediate field deployment for conventional smallholder farmers and long-term stakeholder

engagement. By prioritizing early trust with local smallholders, we aim to establish a reliable commercial foundation that can support sustainable community growth over time.

a. Public & Digital

Digital publications as well as activity on different social platforms is one of our sales strategies. Not only will we reach out to companies, but influencers will also be essential to spread our system. This part will focus on building up credibility for our bioreactor system through the publishing of research papers, which will share our research progress throughout the months as well as results and data to help build trust in the scientific aspect. Furthermore, by launching an official website, information about our product will be more accessible to users, further adding to our credibility. By the second and third quarter, we plan to continue maintaining a digital presence by posting new information as well as anything we feel like would be beneficial for customers/users to know.

b. Industry Outreach / PR

Industry outreach and public relations are essential in order to build credibility and maintain a positive reputation through the media. This piece focuses on establishing ReLEAF's credibility within the agricultural technology and synthetic biology industries before asking smallholder farmers to trust the product. By pitching our containment approach and dual-stress protection as a category first, rather than an improvement on existing biostimulants, we can pursue coverage in agricultural trade publications and outlets. Press releases will also be tied to concrete milestones so that our public narrative is always backed by data rather than just ideas.

c. Conferences & Events

Conferences and events will span across the summer as we use these methods to gain insight from farmers and gain academic insight for our product. Conferences are also an opportunity for experts and farmers to come together. Gaining valuable insights leads to improvements for our system, which could attract more consumers once the changes are made accordingly. Through events, we can better understand the lifestyle of smallholder farmers and how they address stress affecting their crops. Through this valuable feedback, we could adjust our project so that it aligns with their practices.

d. Pilot Programs

By developing pilot programs through a 9-month period in total, we get to let farmers have firsthand experience using our product. Their adaptivity, their acceptance of sustainable agricultural technology, and their critiques are all necessary for us to develop a final product that suits smallholder farmers most. First there would be a 3-month trial during the 2nd quarter, and based on the feedback, we would develop a second stage with a 6-month trial spanning across the 3rd and 4th quarters. With the pilot programs, it gives us insight into raw opinions on our product.

e. Sales and Partnerships

Sales and partnerships are necessary to ensure our product is noticed on the market. Starting with lab sales during the 1st quarter, we start branching out to academic sources, allowing others to gain accessibility to our product. Additionally, since our product relates to agriculture, licensing is necessary. During the 3rd quarter

of our marketing strategy, we would spend these 3 months to ensure that we acquire the licensing needed to be put on the market. Alternatively, we can offer a rental service for farmers wanting to try out our product first before buying it or offer a monthly service for the same reason. We can cater our product to farmers from diverse backgrounds.

f. Educational Outreach

Another one of our sales strategies is to reach more people in agricultural vocational schools, where students there are more likely to benefit from using our product in their future. Our goal is to connect with the future generation of farmers to get them interested in sustainable farming. We plan to achieve this by holding workshops in these vocational schools to promote our technology as well as to instruct these students on how our product works.

g. Regulatory Considerations

Regulation directly shapes ReLEAF's commercialization strategy because the legal classification of the system depends on what is sold, how it is used, and what claims are made. Our initial strategy is therefore to focus on abiotic-stress applications using BoPep4 and ACCD for salinity stress and AtLEA14 for drought stress, while keeping the engineered *B. subtilis* culture contained within the ReLEAF system. In the European Union, this positioning may allow the cell-free outputs to be evaluated under the PFC 6(B) non-microbial plant biostimulant framework for abiotic-stress tolerance, subject to meeting the applicable requirements. In the United States and Taiwan, classification must be determined separately according to the final formulation, intended use, and marketing claims. Consequently, regulatory compliance is a core commercialization requirement that influences ReLEAF's product design, labeling, target markets, and go-to-market timeline. A detailed country-by-country regulatory pathway is provided on the Regulations & Laws page.

6. Risk Analysis

6.1 Risk Assessment & Management

Risk Category	Risk description	Potential impact	Likelihood	Strategy
Technical: Bioreactor membrane permeability	The bioreactor membrane may block LEA peptides, ACCD, and BoPep if the molecular-weight cutoff is too small or if diffusion is slower than expected.	If the protectants cannot pass through the membrane at effective concentrations, the device would fail to deliver the intended plant-protection effect even if the bacteria	Medium	Before Jamboree, we will experimentally test membrane permeability using purified or tagged LEA peptide, ACCD, and BoPep. If the passage is too low, we will test a larger molecular-weight

		successfully produce the compounds.		cutoff membrane or adjust the delivery design while still preventing bacterial escape.
Technical: AND-gate circuit reliability	The high-temperature and high-humidity AND-gate circuit may not activate reliably at the intended environmental thresholds.	The biotic-stress module could release LEA, ACCD, or BoPep too early, too late, or inconsistently, weakening disease protection and increasing waste.	Medium	We will characterize each promoter separately before combining them into the AND-gate system. Thresholds will be tuned in vitro, and activation will be tested under simulated heat and humidity conditions before greenhouse or field testing.
Regulatory: classification of cell-free GMO-derived secretions	Taiwan's regulatory classification of ReLEAF's cell-free secretions, as well as the requirements governing its contained engineered *Bacillus subtilis* culture, remains to be confirmed. The applicable authorities may differ depending on the final formulation. intended use, and marketing claims.	Uncertainty may delay greenhouse or field trials, product registration, commercialization, and partnerships with agricultural cooperatives.	Medium	Seek written guidance from relevant Taiwanese authorities, including the Ministry of Agriculture and APHIA where applicable, and confirm the appropriate agricultural product registration pathway. Keep experiments contained until applicable

				requirements are clarified. Support the assessment with cell-retention validation and tests for viable production organisms and residual recombinant DNA.
Regulatory: unclear pathways in Southeast Asian markets	Requirements for ReLEAF's bioreactor, engineered microbial culture, and cell-free agricultural output may differ across target markets. Existing biosafety and agricultural-product frameworks may apply, but their applicability to ReLEAF's specific system requires country-by-country assessment.	Differences in classification, permits, testing, or registration requirements could delay expansion and increase compliance costs.	Medium	Validate ReLEAF in Taiwan first, then prioritize the Philippines as the first international market, followed by Thailand. Before entry into each country, obtain local regulatory guidance and assess equipment conformity, contained-use biosafety requirements, import rules, and agricultural product registration separately. Expand to subsequent markets only after completing the relevant country-specific review.
Regulatory / Market: organic	The product may not qualify for organic	The organic farming market	Medium	We will not position ReLEAF

certification incompatibility	certification because it is produced using engineered bacteria, even though bacteria are not released into soil.	may be inaccessible, reducing the initial customer base.		as an organic product. Based on the business plan direction, our primary market is conventional smallholder vegetable farmers, who represent the majority of the target market. Our messaging will focus on reduced chemical input, crop protection, and sustainable agriculture rather than organic certification.
Market: farmer distrust of unfamiliar devices	Farmers may prefer familiar sprays over a sealed biological device, especially if the technology seems difficult to understand or maintain.	Slow adoption could limit early sales, even if the product works scientifically.	Medium	We will avoid overly technical marketing language and focus on farmer-facing benefits: fewer spray days, less crop loss during hot and wet seasons, and reduced daily monitoring. Pilot programs with cooperatives and extension offices will generate local testimonial data before broad sales.
Market: slow adoption by cooperatives	Cooperatives, NGOs, and agricultural extension offices may	Distribution could be delayed because direct-to-farmer	Medium	We will prioritize progressive cooperatives and

and extension offices	be cautious about adopting novel agricultural inputs without long-term evidence.	sales are not the most realistic launch pathway.		agricultural organizations already interested in precision agriculture, IoT systems, or sustainable input reduction. First-year pilots will be small, measurable, and designed around clear success metrics such as crop-loss reduction, number of spray events avoided, and farmer satisfaction.
Financial/Market: production cost too high	The cost of the device, cartridge, membrane, and bacterial culture system may exceed what smallholder farmers or cooperatives are willing to pay.	If the price is higher than farmers' seasonal pesticide or fungicide spending, adoption will be limited.	Medium	The financial model will include a sensitivity analysis testing whether ReLEAF remains viable if production cost is 20% higher or price is 20% lower. If individual farmer purchase is not affordable, we will pivot to cooperative purchasing, NGO bulk purchasing, seasonal rental, or cartridge subscription models.
Biological /	The bioreactor	Bacterial leakage	Low-Medium	We will verify the

Regulatory: Bioreactor Membrane Leakage of Live Bacteria	membrane could allow whole *B. subtilis* cells to pass into the environment if the molecular-weight cutoff is miscalibrated, damaged, or degrades over the device's operational lifetime.	into soil would violate containment requirements, risking regulatory rejection and loss of farmer/cooperative trust, regardless of whether the device's functional performance (protectant delivery) is otherwise successful.		membrane's cutoff experimentally against *B. subtilis* cell size and test for degradation over repeated use/simulated field conditions. Membrane integrity checks will be built into a maintenance/replacement schedule, and this containment layer will be validated alongside the genetic and physical layers before implementation into real fields.
Biological / Regulatory: Physical Containment Breach	The device's physical housing could crack, degrade, or be damaged during handling, transport, or extended field use, exposing the internal bacterial compartment directly to soil.	A physical breach would bypass both genetic and membrane containment simultaneously, representing the most direct pathway to environmental release and the most serious regulatory and reputational risk.	Low	We will stress-test the housing for durability under field-like conditions (moisture, temperature cycling, and mechanical handling) and design the compartment to show proof of tampering (if tampered) so that breaches are detectable by farmers/corporatio

				ns. Redundancy with the genetic and membrane layers means a physical breach alone is unlikely to result in bacterial release.

6.2 Legal considerations

Regulatory Strategy and Competent Authorities in Taiwan

Our strategy is intentionally designed around Taiwan's current agricultural legal framework to optimize our time-to-market and ensure strict compliance from the research-and-development (R&D) phase onward. In Taiwan, agricultural inputs are primarily regulated by the Ministry of Agriculture, but jurisdiction is divided based on product claims and targeted mechanisms. Products intended to combat biotic stress, such as pathogens, insects, or weeds, are all classified as chemical pesticides or biopesticides. These fall under the Pesticide Management Act managed by the Animal and Plant Health Inspection Agency. Conversely, products mitigating abiotic stress, such as improving plant growth, nutrient uptake, or environmental tolerance, are governed by the Fertilizer Management Act, which is overseen by the Agriculture and Food Agency (Ministry of Agriculture, 2023).

Product Positioning: The Biostimulant Advantage

We have strategically chosen to position our engineered protectant as a plant biostimulant and fertilizer rather than a biopesticide. Targeting abiotic stress allows us to utilize the fertilizer/plant biostimulant regulatory pathway, which offers significantly simpler regulatory requirements and faster approval times compared to the biopesticide route. While products treating abiotic stress generally carry a lower market value than those targeting biotic stress, this approach drastically reduces initial development bottlenecks and accelerates commercialization.

Requirement and Legalization

To successfully commercialize our product under the regulation of biostimulants, we must obtain the Fertilizer Registration Certificate from the central competent authority before manufacturing the product. Packaging and labeling would be crucial during this stage, as the legal framework has outlined strict regulatory standards by featuring these elements:

- Product Name and Declared Efficacy
- Main Ingredient List
- Product Capacity and Usage Methods
- Storage Conditions and Safety/Environmental Information
- Manufacturer and Production Details

Importantly, our marketing and product must entirely omit any claims of disease control, pest eradication, or biotic stress mitigation. If our product implies pesticidal efficacy, it would immediately trigger regulation of pesticides and undergo a necessary and costly documentation process. In-depth, to register a biopesticide in Taiwan, a company must complete a rigorous, heavily regulated series of tests according to the official "Pesticide Physical and Chemical Properties and Toxicological Test Guidelines" and "Pesticide Field Trial Guidelines." The mandatory requirements include:

- Strain Identification and Biological Characteristics: A comprehensive report detailing the background of the microbial strain's isolation, precise biological identification, and the complete manufacturing process.
- GLP Toxicological and Safety Evaluations: Biopesticides demand extensive Good Laboratory Practice

(GLP) animal safety evaluations. This includes acute toxicity testing (oral, dermal, and respiratory) typically using mammalian models such as rats, as well as tests for eye and skin irritation. Furthermore, environmental safety must be validated through aquatic organism toxicity tests (e.g., using freshwater fish or invertebrates) and non-target insect assessments.

- Efficacy and Phytotoxicity Field Trials: Official field trials must be conducted by government-recognized testing units. These trials require testing across multiple growing seasons to statistically prove pesticidal efficacy, confirm appropriate dosage rates, and ensure no phytotoxic damage occurs to the crops.
- GLP Physicochemical Properties: Detailed reports on the formulation's physical and chemical properties, including long-term storage stability data.

Comparative Timeline and Strategic Rationale

The difference between biostimulant and pesticide regulatory pathways fundamentally affects our time-to-market, capital requirement, and initial experiment design. Choosing biostimulant as our primary goal is a necessary decision for our initial project launch, validated through these comparative analyses:

- Time-to-Market Evaluation: The timeline for fertilizer and biostimulant registration under the Agriculture and Food Agency is highly streamlined. The standard process involves sending a sample to a designated testing unit for a fertilizer specification report (approximately 20 days), followed by document preparation and a government review period of roughly 7 days. In total, a fertilizer registration certificate can be obtained in approximately 30 days. Conversely, registering a microbial biopesticide can take several years due to the mandatory multi-season field trials, lengthy GLP animal testing, and prolonged, multi-phase governmental reviews.
- Capital and Resource Allocation: Biopesticide registration requires heavy upfront capital expenditure to secure GLP-certified laboratories for toxicological testing and authorized agricultural stations for field trials. By positioning our product as a biostimulant, we bypass the need for costly animal testing and multi-season efficacy trials. Our internal R&D can remain tightly focused on optimizing plant physiology—such as measuring ROS accumulation and stress recovery—rather than redirecting funds toward proving mortality rates of pathogens.
- Market Entry Strategy: While biopesticides command a higher market value, the years spent in regulatory limbo represent significant opportunity costs. The biostimulant pathway allows us to commercialize rapidly, generate early revenue, and establish brand trust with farmers through measurable improvements in abiotic stress tolerance.

7. Financial Projection

7.1 Revenue-Cost Analysis

Our cost and price depend on the scenario: the farm size, the crop types, and the plant density all change how many bioreactors, sensors, and cartridges a farm needs. Modelling every combination would give a large grid of results that is hard to read and plan around. To simplify, we calculate everything per hectare, because dose, hardware, and cartridges all grow with the area planted.

Farms in Taiwan differ a lot in size, and a model built on farms would need a separate scenario for every size. When we tested sizes from 0.5 to 20 ha, profit per hectare stayed close to US$2,000 to 2,500. Counting hectares therefore gives one break-even target (about 230 ha), however many farms they come from, and it matches how we would sell: by area covered, including through cooperatives. Farm size still matters for how many customers we must win, and the sections below show that effect.

We cost one reference farm: 1 ha, the average Taiwanese farm (Liu, 2018), planted with 3 crop types at 22,000 plants per ha. Everything is calculated per hectare, so one break-even target covers any mix of farm sizes. Year 0 is a build year with fixed costs and no sales; years 1 to 5 are selling years.

Assumptions

The model uses one reference farm and the settings below.

Assumption	Value
Market	Taiwan only, Years 0 to 5.
Reference farm	1 ha, 3 crop types, 22,000 plants per ha, 9 sprays in a 10-month season
Package	2 bioreactors (1 L each), 3 moisture sensors (one per crop type), 1 refill cartridge (100 g) a year
Pricing	Price to the farmer = 2 times our full cost (parts, warranty reserve, freight, setup visits, expired stock)
Team	3 people in Years 0 and 1, 4 in Years 2 and 3, and 5 in Years 4 and 5. US$1,500 a month for 14 months, plus 20% on costs
Not counted	Interest, resale value, inflation, and product registration. Tax (20%, unconfirmed) is shown only as a separate estimate Medium sold to farmers who choose not to make it from hay. We cannot yet predict how many will buy, so the base case counts no medium revenue or cost.

Fixed cost

Fixed cost is about US $147,000 *in Year 0 and US* $627,000 across years 0 to 4. Year 0 carries the one-off machines and R&D (about US$53000). It falls to US$94,000 in Year 1, then rises as the team grows from 3 to 5 people.

Fixed cost, US$	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Staff	79,000	79,000	106,000	106,000	132,000	132,000
Running costs (rent, power, insurance, accounting, marketing)	14,000	14,000	14,000	14,000	14,000	14,000
Machines and R&D	53,000	0	0	0	0	0
Total	**147,000**	**94,000**	**120,000**	**120,000**	**147,000**	**147,000**
Team size	3	3	4	4	5	5

Variable cost and price per hectare

A 1 ha farm pays about US$ 5500 in Year 1 for a package that costs us about US$ 2800, which leaves about US$ 2800 of profit. In later years it pays about $US90 for a refill cartridge, which leaves about US$45.

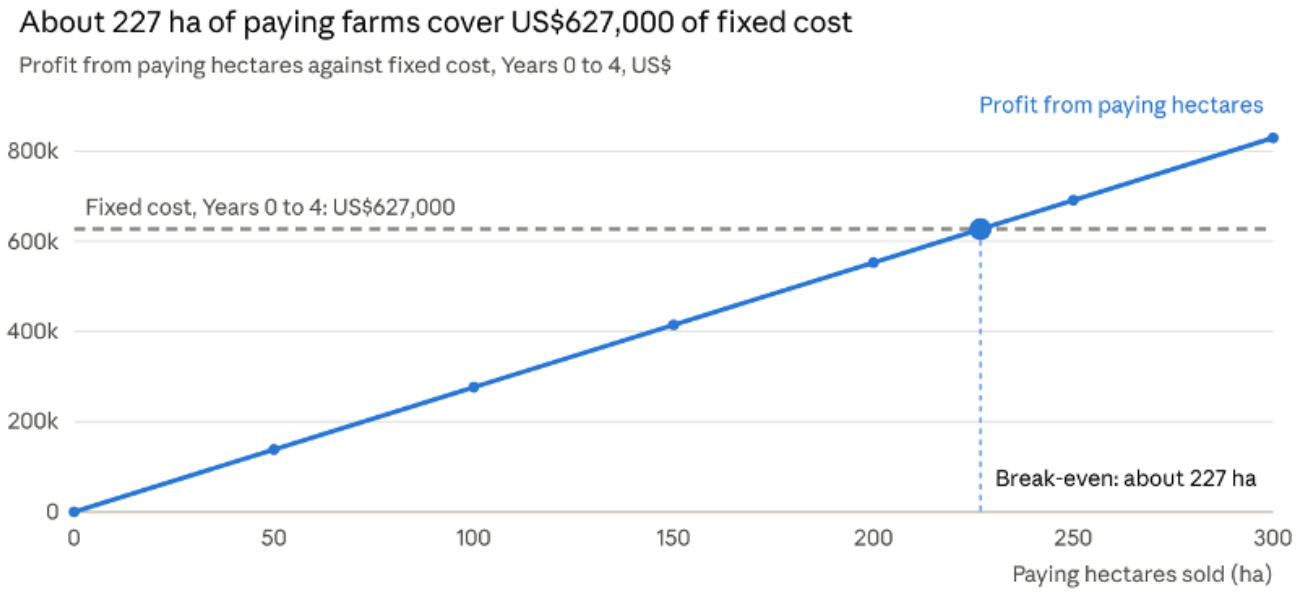
Item (Year 1)	Quantity	Our cost, US$	Price to farmer, US$
Bioreactors	2	2,280	4,550
Moisture sensors	3	190	380
First refill cartridge	1	45	90
Freight, set-up visits, warranty		260	510
Total		**2,800**	**5,500**

Break-even analysis

We need to sell about 230 ha of paying farms in Years 0 to 4 to cover US$627,000 of fixed costs.

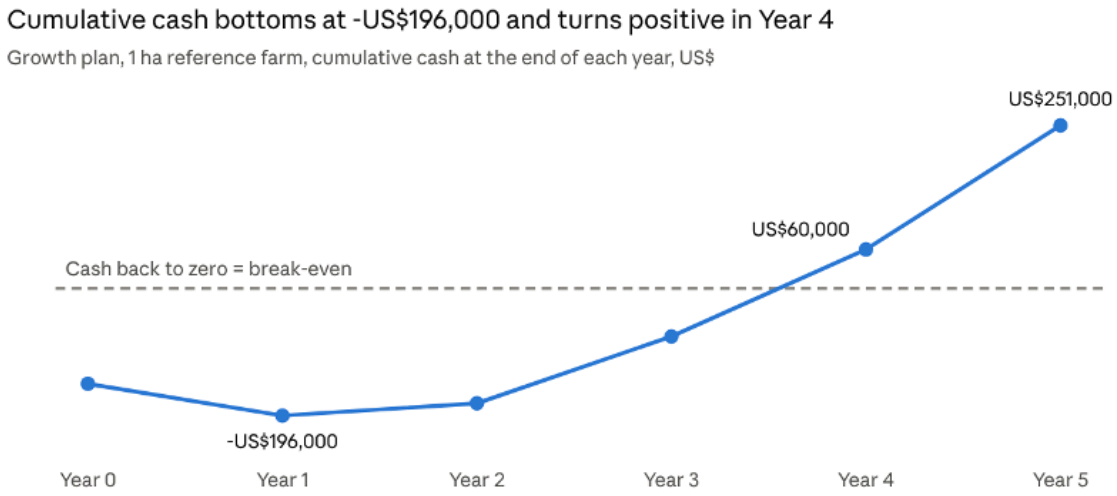
$$\text{hectares to break even} = \frac{\text{fixed cost, Years 0 to 4}}{\text{profit per hectare}} = \frac{627{,}000}{2{,}765} \approx 227$$

At 1 ha per farm, that is about 230 farms, or about 57 a year over years 1 to 4. The figure is cautious: it ignores renewal profit and the cost of the free pilot.



Growth plan

Under the growth plan, cumulative cash bottoms out at -US$196,000 in Year 1 and turns positive in Year 4. The plan sells 20 paying hectares in Year 1, before field results exist.



Growth plan	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Paying hectares sold	0 (build year)	20 (+4 free pilot)	50	80	100	120
Team size	3	3	4	4	5	5
Refill cartridges sold	0	24	67	127	189	252
Fixed cost (US$)	147,000	94,000	120,000	120,000	147,000	147,000
Profit before fixed cost (US$)	0	44,000	139,000	223,000	280,000	338,000
Cumulative cash (US$)	**-147,000**	**-196,000**	**-177,000**	**-74,000**	**60,000**	**251,000**

Return for investors

If we hit the growth plan, investors who fund the US$196,000 low point get 2.3 times their money back by Year 5, an IRR of 24%.

Measure (Year 0 to Year 5, no resale value)	Growth plan	Conservative case
Funding needed (US$)	196,000	236,000
IRR before tax	24%	-7%
IRR after 20% tax (estimate)	21%	-7%
Money back by Year 5, before tax	2.3 times	0.7 times
Money back by Year 5, after tax	2.0 times	0.7 times
Break-even year	Year 4	Year 6
Payback, counted from Year 1	3.6 years	Not within 5 years
Payback, counted from Year 0	4.6 years	Not within 5 years
NPV at 10% discount rate (US$)	111,000	-109,000
NPV at 25% discount rate (US$)	-4,000	-146,000

The IRR includes the Year 0 build-year cost. Money back = (funding + cumulative cash in Year 5) ÷ funding. The 20% tax rate and loss carry-forward are unconfirmed

Summary

To cover Years 0 to 4 fixed cost of about US$ 627,000 we must sell about 230 ha to paying farms, at roughly US$2,800 profit per hectare.

- Package: about US$ 5,500 in Year 1 (2 times our cost) and US $ 90 a year for refills.
- Growth plan: break-even in Year 4, peak funding about US$196,000, IRR 24% before tax (21% after a 20% tax estimate), 2.3 times money back by Year 5.
- Conservative case (70% of planned hectares): break-even in Year 6, funding about US$236,000, IRR about -7%.
- Main risks: the product is not yet tested on any crop, and the farmer must save about US$1,900 a year to pay back in 3 years.

Detailed calculation shows in: ReLEAF Business Model Calculator

7.2 Investments

For our funding, we decided to develop it into three development stages: R&D to iGEM Jamboree, post-iGEM proof-of-concept validation, and first commercial pilot deployment.

Stage 1—R&D to iGEM Jamboree
Our primary goals are building a functional prototype system, validating peptide expression and secretion, characterizing AND-gate responses, demonstrating bioreactor feasibility, and generating preliminary safety and efficacy data. The sources that will likely fund us are school/university innovation funding, small fundings from biotech companies, and Taiwanese education grants. When people invest in a project, there are essential aspects that they are looking for. Therefore, our project needs early proof of concept data, HP and stakeholder engagement, and a distinct innovation compared to existing agricultural technology.

Stage	Amount ($USD)	Use of Funds	Source (named program)	Milestone
Stage 1: Proof of Concept and Early Validation	$4,600-$9200	Lab consumables, gene synthesis, reagents, growth media, and stakeholder interview travel/logistics.	University Student Innovation / Venture Grants	Successful secretion of LEA peptide. Initial stakeholder interviews (farmers, agricultural extension officers)
Stage 2: Technical Feasibility & System	$154000– 30,800	Specialized membrane material	Small Biotech Corporate	Characterizing AND-gate circuit

Validation		testing, environmental chamber access, and initial bio-safety observations.	Sponsorship	responses under environmental triggers Preliminary crop safety observations
Stage 3: Local Commercialization & Compliance Prep	$30,800	Pilot-scale testing, local crop safety trial protocols, regulatory preparation, and commercial vendor setup.	Local Government Subsidy	Validation of system performance in field-simulated settings Completion of the HP/stakeholder alignment report for regulatory filing
Stage 4: Market Entry & Supplier Onboarding	$192,600 – $195,700	Seed round of about US$ 196,000 in two tranches: about US$144,800 in Year 0 for a 3-person team, production systems and R&D and running costs; about US$46K in Year 1 for a free pilot on 4 ha and costs not yet covered by sales. Plus vendor registration fees, platform listing costs, product compliance filing, and administrative processing fees (US$1,540 – $4600).	Agricultural biotech investors / climate-tech funds (seed round); Corporate Angel Fund / Incubator Commercialization Fund	Pilot passes its threshold; first 20 paying hectares sold. Supplier Registration Completion

With these needs in mind, successful secretion of the LEA peptide, initial bioreactor membrane compatibility testing, environmental trigger activation data for the AND-gate circuit, basic crop safety observations, and

stakeholder interviews are all key milestones linked to funding.

Stage 2 – Post-iGEM proof-of-concept validation
For us, our project is not just for the Jamboree; we want to develop our system into a long-lasting product that can help smallholder farmers across the world. A few goals for our project include a transition from a laboratory prototype to a field-relevant system, conducting small-scale crop trials, validating efficacy under combined abiotic and biotic stress, and improving stability and manufacturability. When requesting funding, our team recognizes aspects that funders need to see. These aspects include safety validation, evidence of farmer demand, cost feasibility analysis, and evidence of crop benefit. Funding sources post-iGEM include Taiwan Ministry of Agriculture innovation programs and other agricultural programs.

Stage 3 — First commercial pilot
Our goals are to deploy pilot systems with real farmers to test adoption and usability, validating seasonal economic viability, and preparing for regulatory approval pathways. Some likely funding sources for our first commercial pilot include agricultural biotechnology investors, climate tech venture funds, and sustainable agriculture accelerators. In this case again, we address that funders need to see strong field trial data, a clear regulatory pathway, a scalable manufacturing plan, evidence of farmer adoption, and a defined target market.

8 Growth Potential

8.1 Growth Potential

8.1.1 market expansion

Year 1: Taiwan (Home Market)
We will first launch our product in Taiwan because we already have local partnerships with farmers and agricultural organizations, making testing and communication easier. We are also familiar with local regulations and can collect field data under real farming conditions. Before expanding, we will complete biosafety approvals, conduct our pilot programs, and demonstrate that our system reduces crop losses and chemical use while receiving positive feedback from farmers.

Year 2: Philippines
After success in Taiwan, we will expand to the Philippines, where many vegetable farms face heat stress, flooding, and crop diseases. Before entering the market, we will complete local regulatory approvals and field testing. Expansion will begin once we have proven success in Taiwan, published field trial results, found a local partner, and have enough production capacity to support pilot programs.

Year 3: Vietnam
Vietnam will be our next market because it has a large vegetable farming industry and is investing more in agricultural technology. We will obtain the necessary biosafety and environmental approvals and test our system under local conditions. Expansion will take place after a successful pilot in the Philippines, a local distribution partner, and proof that the system works well in different climates.

Year 4: Southeast Asia
After building a strong presence in Taiwan, the Philippines, and Vietnam, we will expand to other Southeast Asian countries such as Thailand and Indonesia. These markets require more regulatory approvals and local testing, so we will only enter after achieving commercial success in at least three countries, increasing our production capacity, and showing that our system performs well across different crops and growing conditions.

Year 5+: Europe and USA
Moving the market of our bioreactor in the European Union and the United States will be beneficial in both the sales direction and the awareness direction. For expansion into Europe, a relatively short registration process is needed to gain market access to the entire EU, i.e., non-necessitating the need to pass regulatory gates of each individual country. In addition, the EU has previously, under the regulations 2019/1009 PFC6(B) for BoPep4, established non-microbial biostimulants, which is similar to the definition of ReLEAF's biostimulant, therefore making it easier to defend our claim. In the United States, our protectant would most likely be described as a beneficial substance for the soil-plant environment, and since the bioreactor protects against salinity and heat stress, which are environmental factors, it will be strictly named a "abiotic stress tolerance" product, which avoids directly implying the biostimulant on the plant.

8.1.2 Product extension

Potential product variations:

- Drought and soil pathogen protection
- Salinity and bacterial disease defense
- Heat and virus vector control
- Rice growth support
- Different potential target crops (such as strawberries and tomatoes)
- Solely selling a bioreactor

Why choose these?

- Drought and soil pathogen protection is necessary for planting rice
    - For most of the population of Asia, rice is a carbohydrate that routinely appears in the daily meals of individuals. It may also be known as the most widely consumed food overall. Due to these high demands, rice is required to be harvested in large amounts and up to 2.0×10^9 tonnes by the year of 2030. However, drought has been occurring in the key harvesting areas, and the lack of water irrigation systems along with scarce rainwater deeply affects rice production beginning from the germination stage. Seed germination of rice requires specific soil humidity, but droughts create an unstable soil environment as well as inhibition of water uptake (Panda et al., 2021).
    - Soil pathogens play a large role in decreasing the crop yield of many crops other than just rice. It works by attacking roots, stems, and leaves at the same time, allowing crop losses even up to 70% in severe cases. A way that some soil pathogens work is destroying root systems, preventing

plants from absorbing nutrients and water from the soil, leading them to die or wilt from malnutrition. Seed rot may also lead to weak or overly thin seedlings, lowering plant density and lowering yield early in the growth season (Liang et al., 2023).

- Salinity and bacterial disease defense are necessary for coastal farming
    - Soil salinization is an escalating abiotic threat that is driven by rising sea levels and over-extraction of groundwater. When salt-water infiltrates agricultural lands, it increases the osmotic pressure of the soil and makes it difficult for the crops to absorb water and essential nutrients. It creates a state of "drought" in the plant, since there is water all around, but water is still being pulled out of the plant itself. It ultimately leads to plant wilting and causes terrible damage to the overall crop yield (Md Isfatuzzaman Bhuyan et al., 2024).
- Heat and virus vector control is necessary for all types of crop growth
    - Rising global temperature has increasingly presented itself as a larger and larger problem in the agricultural world. Extreme heat alters the plant's ability to survive and disrupts photosynthesis, damaging cell membranes and weakening the plant's natural immune response. Plants then are more prone to environmental degradation, especially when they face heat stress early in their growth cycle. During flowering stages, it could even lead to pollen sterility, stunted development, and, commonly, yield declines (Samat et al., 2025).
    - Insects such as whiteflies and aphids can often act as a transportation mechanism for plant diseases and viruses. In addition, rising temperatures can also accelerate their reproductive life cycles, expanding their range and increasing their impact across large acres of land. As they feed on crops, they transmit pathogens that impact the plant's own systems, causing wilting or death and therefore decreasing the yield drastically. Controlling these is crucial to preventing the spread of pathogens and viruses through entire fields of crops (Trebicki, 2020).
- Rice growth support
    - Rice is highly sensitive to salinity, particularly during the early seedling and reproductive stage (Ullah et al., 2022) (Prusty et al., 2018). Unlike wheat, rice has a poor Na^+ exclusion pathway in its roots. Salt travels quickly and causes premature leaf senescence of older leaves and severe stunting. Salt accumulation in flower tissues disrupt the growth of microsporogenesis, leading to high rates of spikelet sterility (Hussain et al., 2024).
    - When soil moisture drops, rice synthesizes abscisic acid (ABA) in the roots, which travels to the leaves to trigger stomatal closure. This minimizes water loss through transpiration but cuts down CO_2 intake, decreasing the photosynthetic rate (Bhandari et al., 2023). Reduced carbon intake results in excess light energy in the chloroplasts, causing the overproduction of Reactive Oxygen Species (ROS) such as H_2O_2 and OH^-. These ROS molecules oxidize and degrade cell membranes, causing electrolyte leakage and cellular collapse. (Jarin et al., 2024)
- Different potential target crops (such as strawberries and tomatoes)
    - Strawberry is ranked as the most salt-sensitive fruit crop in major salt tolerance databases, and it is especially vulnerable to chloride. Once soil salinity passes a low threshold, yield drops about 33% for each additional conductivity unit (Ferreira et al., 2019). Chloride builds up most in the roots, then leaves, petioles, and fruit, and marginal leaf burn is a visible sign of chloride toxicity.

This shows strawberries suffer from both the osmotic stress and the specific ion toxicity of salt (Grieve et al., 2010). Excess Cl^- disrupts leaf metabolism, reducing photosynthesis and carbohydrate production, which lowers both fruit yield and quality. Importantly, yield loss begins well before any leaf symptoms appear (Haifa Group, n.d.). This makes strawberries a strong fit for ReLEAF, since farmers can't rely on visual cues to know when to intervene.

    - Tomato is drought-sensitive, and water stress lowers its growth, yield, and fruit quality while raising disease risk (Mahapatra et al., 2025). Similar to rice, drought triggers ABA-driven stomatal closure in tomato, and ABA works together with ROS to close the stomata (Xu et al., 2022). This conserves water but limits CO_2 uptake and photosynthesis. Water and salt stress also damage the fruit directly through blossom-end rot (BER), a physiological disorder tied to disrupted calcium (Ca^{2+}) balance and irregular watering. ROS accumulation, combined with Ca^{2+} imbalance, strongly increases how severe BER becomes (Ferrari et al., 2010). Since BER makes fruit unmarketable, protecting tomatoes from abiotic stress directly protects farmer income.
- Selling a bioreactor that can dispense market protectants by itself for farmers who are against GMOs in their crops and ease the product by establishing trust beforehand.
    - According to the answers of countless farmers at last time's Taiwan Farmer Expo interviews, most farmers are against GMOs and would not like something that is unnatural to "infect" their crops. In fact, when you approach their stalls, most farmers boast that their crops are non-GMO and completely safe for adult and children's consumption. This complicates the launch of our product, as the market base may be smaller if the only variation in the product is targeting different types of stress. However, selling the bioreactor on its own would be another way to speed up the farming process and make it easier on farmers while respecting their wishes to keep their agriculture non-GMO.
    - Another largely faced issue in the agricultural field is the aging farming population. No more farmers want to harvest food because of the long hours, lack of experience, and life in rural areas. With our product and the technology that comes with it, it could help the next generation of farmers to realize that agriculture does not have to be as difficult as they last remembered, solving the issues of lack of experience and long hours, since the bioreactor does all the work and is easier to use than manually checking on the plants daily.

Why do these matter?

- The crops targeted by our protectant and bioreactor have been filtered through both Taiwanese data and global data. In 2024-2025, the crop cycle produced 1.087 million metric tons of rice and 235,000 hectares of harvested area (FAS USDA). However, the efficiency has decreased, shifting from only 5.27 metric tonnes per hectare to 4.6 metric tonnes per hectare from 2016 to 2025, solely in the rice harvesting in Taiwan. Shifting to a global perspective, over 75% of global rice is cultivated under flooded conditions; however, the demand for rice increases as the global population demonstrates an exponential upward growth in the global population. Rice is the primary carbohydrate source for an estimate of 557 million people across Asia alone, and the global rice consumption is projected to grow at an annual rate of 0.9% from 2023 to 2032 (Shahzad et al., 2025).
- Using a dual-module platform in our product increases the benefits and allows the product to be spread to a larger market base. It would be framed and sold as the same project and model, but the contents

would change to target different types of agricultural conditions and crops. They are important to the health of each crop, and if there is a possibility to combine two protectants into one, it would be more efficient and easier for farmers to use and conduct, using only one product to increase the health of their crops.
- Ideally, the farmers would change their mind towards GMOs once they have tried our product, but by forcing the GMO-based product onto farmers, it could decrease reliability and trust, even if it benefits their yield. By first using a bioreactor that reduces work load and hours on the field that is compatible with protectants they are already using, it would establish trust that could later persuade them into using our GMO product.

Possible challenges

- The different parts made to target each stress factor could have negative or any interactions with each other when placed in the same module of the bioreactor. In addition, there is no confirmation of how each product would react when mixed together and then released into the soil. There may even be a possibility where the protectants become too overwhelming for the crops and create a negative effect on the yield instead of increasing it.
- The form in which their protectant or fertilizer comes in varies, like in liquid, powder, or gaseous forms, which would be implemented into the bioreactor in different ways. In the case where the protectant could come in different states of matter, we would need to find a way for the bioreactor to either accommodate the protectant in all states or for the state to be converted into a central one before it can be diffused into the land.

8.2 Exit Strategy

- Exit Strategy 1: Acquisition by a company

Acquisition by an agrochemical or agricultural biotech company (e.g., BASF, Corteva, or Bayer) is the most likely scenario. Acquisition can enable these companies to use our engineered *Bacillus subtilis*, hardware, gene circuit, and customized system model. With the data we've collected and the system we developed, they can further develop and expand to their current products and business model. For the investors, this could be a low-risk exit route.

- Exit Strategy 2: Regional Licensing to Local Manufacturers

A lower capital and faster way is licensing the gene circuit and system to local manufacturers in each target market (e.g., Taiwan, the Philippines, and beyond) instead of manufacturing and distributing by ourselves. This can avoid the large amounts of funds caused by building the contract manufacturer in each territory. Let the local partner, who already understands the regional supply chain, agricultural distribution networks, and their country's specific biosafety approval processes, speed up the whole process more efficiently. Also, this path is particularly suited to markets that need to gain their regulatory trust by getting approved by their government sector.

- Exit Strategy 3: No-exit Social Enterprise Model

The third path is to continue operating as a social enterprise, reinvesting the operating surplus into farmer subsidy programs rather than returning capital through acquisition. This option is the most direct way to show our continuity of the mission established in our Social Business Model Canvas (Section 3.1), where the paying customer is explicitly distinguished from the intended beneficiary, which indicates that the surplus they receive is often different. Through this path, the surplus revenue would be used to expand rental ownership access for lower-income farmers, effectively making this exit a transition into a sustaining nonprofit.

KPIs

Phase 0: Concept Validation (Year 1)

- Units Deployed: 0 (Strictly laboratory-scale testing using liquid media and lab soil environments).
- Technical & Biological Metrics:
    - Achieve high gene transformation efficiency in *Bacillus subtilis* models.
    - Successfully trigger protectants (LEA 14 and ACC deaminase) using green (535 nm) and red (670 nm) light.
    - Mitigate salinity stress in *Arabidopsis thaliana* plant models at salt levels of 100-150 mM NaCl.
    - Demonstrate reversible, light-controlled biological secretion inside a sealed container by the iGEM Jamboree.
- Strategic Metrics: * File intellectual property (IP) patents for engineered strains and light circuits.
    - Build data models mapping vulnerabilities for target crops (rice, vegetables, coastal crops).

Phase 1: Legal Navigation & Field SOPs (Year 1 – Year 2)

- Units Deployed: 20–50 prototype devices in local validation fields
- Farmers Reached: 20–70 smallholders and research farm workers.
- Financial & Business Metrics:
    - Revenue: Low (primary focus is on testing reliability, not generating profit).
    - Production Cost: High initial per-unit costs.
    - Target Pricing: Cartridge refills must cost equal to or less than a farmer's standard seasonal chemical fungicide expenses.
    - Adoption Strategy: Introduce the physical hardware first to build operational trust; introduce the bio-cartridges later.
    - Business Model: Roll out (primary seasonal rentals and shared ownership options to help lower financial risks for smallholders.
- Regulatory & Validation Metrics:
    - Legally prove "absolute containment" (using 0.22 μm filters) to Taiwan regulators (BAPHIQ & MOA) to bypass strict environmental GMO laws.
- Social Metrics:
    - Fungicide Spray Events: 20–30% fewer chemical sprays per season.
    - Crop Loss Reduction: 10–15% improvement in crop yields during bad weather or disease stress.
    - Labor Reduction: 10–20% less time spent by farmers manually walking fields to check on crops.
    - User Experience (UX) Score: Over 70% positive feedback on the simple, single-button physical

design and high-contrast mobile app icons.
    - Customer Renewal Rate: 20–30% of pilot farmers choose to continue using the system after the first trial.

Phase 2: Industrial-Scale Readiness & Global Market Integration (Year 2 – Year 4)

- Units Deployed: 150–400 units deployed across regional farming cooperatives worldwide.
- Farmers Reached: 500–1,500 farmers.
- Financial & Commercial Metrics:
    - Revenue: Early commercial revenue streams begin.
    - Production Cost: Reduced by 30–40% through manufacturing optimization.
    - Business Model Shift: Transition from direct hardware sales to technology licensing with large distributors.
    - Data Monetization: Activate a secondary revenue stream by selling anonymized, real-time climate/soil data to external market researchers while using it internally to optimize products.
    - Customer Renewal Rate: 50–60% of users renew their service contracts.
- Engineering & Scale Metrics:
    - Redesign the bioreactor to handle automated cooling, oxygen mixing, and heat removal to stop toxic waste buildup during mass production.
    - Successfully engineer stress-tolerant protectants that survive harsh field transport.
- Expansion into the global market:
    - Form pilot testing partnerships with the International Rice Research Institute (IRRI) and regional state universities.
    - Deploy salt-mitigating units in Vietnam's Mekong Delta and high-density vegetable areas (like Da Lat) using the streamlined approvals under Decree 43/2026/ND-CP.
- Social & Agronomic Metrics:
    - Fungicide Spray Events: 30–40% decrease in chemical applications.
    - Crop Loss Reduction: 15–25% yield improvement with much more consistent harvest volumes across changing seasons.
    - Chemical Input Reduction: 20–30% lower total chemical use.
    - Farmer Income Improvement: 10–20% average increase in seasonal income due to lower chemical costs and saved crops.

Phase 3: Global Infrastructure & Multi-Market Diversification (Year 4 – Year 5)

- Units Deployed: 500–1,500 units working actively across 2–3 international territories.
- Farmers Reached: 3,000–10,000 farmers (mainly through large regional farming cooperatives).
- Expansion into Global Market: Expand into markets such as Southeast Asia in places like Indonesia as well as the US and places in Europe
- Financial & Production Metrics:
    - Revenue: Established, stable global commercial scale with multi-market income streams.
    - Production Cost: Reduced by 50–60% compared to Year 1 by shifting from local assembly to global contract manufacturing.

    - Customer Renewal Rate: 60–75% customer retention.
- Market Diversification Metrics:
    - Bundle A (Complete Solution): Successful distribution of the pre-packed bioreactor + bio-cartridges to high-yield corporate farms in progressive regulatory zones.
    - Bundle B (Hardware & API): Successful sales of the standalone hardware + open software API + universal growth media, allowing research labs and global corporations to program the device to grow their own local, native microbes.
- Social & Environmental Impact Metrics:
    - Fungicide Spray Events: 40–50% fewer total applications per season.
    - Crop Loss Reduction: 20–35% protection improvement during extreme, stress-heavy weather seasons.
    - Chemical Reduction: 30–50% lower chemical pesticide use overall, leading to a measurable drop in local soil toxicity.
    - Farmer Income Improvement: 15–30% increase in seasonal profits.
    - Community Resilience: Proven reduction in climate-driven debt cycles for aging smallholder families, preserving ancestral land assets for the next generation.